

Week3_Sun_802_Quantum Computing_Lecture_Note

Quantum Computing 课堂笔记与翻译

Quantum Computing Lecture Notes - Level 9 Master of Software Engineering

Course Administration & Homework Reminder

HOMEWORK CHECK AND SUBMISSION INSTRUCTIONS

- The lecturer reminded students about a previously assigned task.
- Some students have shared their work, but not all.
- **Submission method:** Work can be shared on the Teams platform.
- **Location:** Files are available under **Week 2** on Teams (Share → Files → Week 2).
- **Task:** Complete the first notebook exercise, including all reflections.
- Students who have not finished are instructed to do so.

 **Student:** How can I share, like...  **Lecturer:** You can share it on the team, so you can just do...

 **Student:** What's the homework? What did he ask for?  **Lecturer:** [Referring to another student] What did he ask for?  **Student 2:** I just remember his. The gate. Remember that? And the X gate and the gate.

Lecture Recap: Previous Class Content

REVIEW OF LAST SESSION

The lecturer began by asking students to recall what was covered in the previous class.

 **Lecturer:** So we'll start with the recap what we have done last time. So anyone can help me recalling what we have done in our last class?

Topics covered in the previous session:

- **Bloch sphere** - discussed in detail
- **Quantum gates** studied:
 - **Hadamard gate** (Edmund gate)
 - **X gate**
 - **Y gate**
 - **Z gate**
 - Other gates discussed
- **Complex method**
- **Entanglement** - example discussed
- **Superposition** - exercise around superposition and entanglement

 **Lecturer:** So we did, I think, the exercise around superpositioning and entanglement. So to understand it.

Today's Lecture Plan

LEARNING OBJECTIVES

The lecturer outlined the plan for today's session:

1. **Understand number systems** - specifically **complex numbers**
2. **State vector analysis:**
 - What information can be revealed from a state vector
 - How to read state vectors
 - How to plot state vectors
 - Practical exercise to understand state vectors
3. **Bloch sphere** - introduction and deeper understanding

IMPORTANCE OF COMPLEX NUMBERS

 **Lecturer:** But main idea is to understand complex numbers. Those who understand, they have a just here with me. But I think in majority, most of you will appreciate the understanding of complex number, because that's what we need in future, to understand complex numbers.

Core Academic Content: Number Systems

WHAT IS A NUMBER SYSTEM?

The lecturer initiated a discussion on number systems:

 **Lecturer:** What are complex numbers? Like what is any number system per se? What is number system?

 **Student:** Binary number system, okay.  **Lecturer:** Any other number system? 
Student: Decimal number system.

中文全文翻译

量子计算课程笔记 - 软件工程硕士 Level 9

课程管理与作业提醒

作业检查与提交说明

- 讲师提醒学生关于之前布置的任务。
- 部分学生已分享他们的作业，但并非所有学生都完成了。
- **提交方式：**作业可以在 Teams 平台上分享。
- **文件位置：**相关文件位于 **第2周** 文件夹下（Teams → 共享 → 文件 → 第2周）。
- **任务内容：**完成第一个笔记本练习，包括所有反思部分。
- 尚未完成的学生被要求补做。

 **学生：**我该怎么分享，比如...  **讲师：**你可以在团队里分享，直接操作就行...

 **学生：**作业是什么？他要求了什么？  **讲师：**[指向另一位学生] 他要求了什么？ 
学生2：我只记得他的。那个门。还记得吗？还有 X 门和那个门。

课程回顾：上节课内容

上节课复习

讲师首先请学生回忆上节课的内容。

 讲师：我们先回顾一下上节课的内容。谁能帮我回忆一下我们上次课讲了什么？

上节课涵盖的主题：

- 布洛赫球 - 进行了详细讨论
- 量子门 学习内容：
 - 哈达玛门
 - X 门
 - Y 门
 - Z 门
 - 讨论的其他门
- 复数方法
- 纠缠 - 讨论了相关示例
- 叠加态 - 围绕叠加和纠缠进行了练习

 讲师：我想我们做了关于叠加和纠缠的练习，目的是理解这些概念。

今日课程计划

学习目标

讲师概述了今天的课程安排：

1. 理解数制 - 特别是复数
2. 状态向量分析：
 - 从状态向量中可以揭示哪些信息
 - 如何解读状态向量
 - 如何绘制状态向量
 - 通过实践练习理解状态向量
3. 布洛赫球 - 介绍与深入理解

复数的重要性

 讲师：但主要目的是理解复数。已经理解的同学可能觉得很简单。但我认为大多数同学会感谢对复数的理解，因为这是我们未来需要的基础。

核心学术内容：数制

什么是数制？

讲师发起了关于数制的讨论：

👤 讲师：什么是复数？或者说，什么是数制本身？什么是数制？

👤 学生：二进制数制，好的。👤 讲师：还有其他数制吗？👤 学生：十进制数制。

Core Academic Content: Number Systems

What is a Number System?

The lecturer continues the discussion on number systems:

👤 **Lecturer:** What are complex numbers? Like, what is any number system per se? What is a number system?

👤 **Student:** Binary number system. 👤 **Lecturer:** Okay. Any other number system? 👤 **Student:** Decimal number system.

👤 **Lecturer:** Okay, before we go, we'll discuss in brief. So, we need to understand complex numbers. I asked you what a number system is, and you gave me some examples. Now I'm asking you: everyone is familiar at least with what *linear algebra* is? *Algebraic equations*? Everyone is kind of familiar with algebraic equations. So, whatever the number system you know, can you solve this equation for me?

The Motivating Equation: ($x^2 + 1 = 0$)

👤 **Lecturer:** The equation is ($x^2 + 1 = 0$). This is a quadratic equation. And whatever the number system you think of, can you solve this? So how to solve it? So ($x^2 = -1$). And what is the square root of -1?

👤 **Student:** Complex.

👤 **Lecturer:** Complex. What is this? For those who don't want to dig more into mathematics, the point is: you take any number. We are going to solve an equation where we will take the solution of X. Whatever number we put in, whether it is negative or positive, when you square it, what will happen? Whenever you square any number, even

a negative number, it will become positive. So this is always going to be positive. That means it will always be greater than 0. Now, anything greater than 0, you add it to 1, that will always be greater than 0. So how is it possible that it will be equal to 0? To make it equal to 0, this number (x^2) should be equal to -1. Only then is it possible that this equation gets satisfied.

The Philosophical Necessity of Complex Numbers

 **Lecturer:** And when we can write an equation like this, that means definitely in nature, or somewhere, this kind of a problem can exist. You can reflect any problem, or you can sort of model any problem through this equation. We don't know in nature what follows this kind of a pattern. We don't know. But when it comes to our mind, it is a possibility. Like, there must be something in nature which may follow this kind of modeling: ($x^2 + 1 = 0$). x could be anything. I don't know what that x is. But in nature, when I can write this equation like this, something could be modeled out of this equation: ($x^2 + 1 = 0$).

The Evolution of Number Systems

 **Lecturer:** Now the point is, whatever the systems you talked about, we will see some kind of numbers. The numbers could be positive. So, everyone agrees to it? How many types of numbers could be there? Positive numbers. Why? If we go back to ages, maybe 10,000 years before, whenever this number system might have come up, why would there be...

中文全文翻译

核心学术内容：数制

什么是数制？

讲师继续讨论数制：

 **讲师：**什么是复数？或者说，什么是数制本身？什么是数制？

 **学生：**二进制数制。  **讲师：**好的。还有其他数制吗？  **学生：**十进制数制。

 **讲师：**好的，在我们继续之前，我们简要讨论一下。所以，我们需要理解复数。我问过你什么是数制，你给出了一些例子。现在我要问你：每个人都至少熟悉什么是线性代数

吗？代数方程？大家多少都熟悉代数方程。那么，无论你知道什么数制，你能为我解这个方程吗？

动机方程：($x^2 + 1 = 0$)

👤 讲师：方程是 ($x^2 + 1 = 0$)。这是一个二次方程。无论你想到什么数制，你能解这个方程吗？那么如何解它？所以 ($x^2 = -1$)。 -1的平方根是什么？

👤 学生：复数。

👤 讲师：复数。这是什么？对于那些不想深入钻研数学的人，关键是：你取任何数。我们要解一个方程，我们要找到X的解。无论我们代入什么数，无论是负数还是正数，当你对它平方时，会发生什么？当你以任何数（即使是负数）进行平方时，它都会变成正数。所以这总是正数。这意味着它总是大于0。现在，任何大于0的数，你把它加1，结果总是大于0。那么它怎么可能等于0呢？要让它等于0，这个数 (x^2) 必须等于-1。只有这样，这个方程才有可能被满足。

复数的哲学必要性

👤 讲师：当我们能写出这样的方程时，这意味着在自然界中，或者在某个地方，这种问题一定可以存在。你可以通过这个方程反映任何问题，或者对任何问题进行建模。我们不知道自然界中有什么遵循这种模式。我们不知道。但当我们思考时，这是一种可能性。比如，自然界中一定存在某种东西可能遵循这种建模： ($x^2 + 1 = 0$)。X可以是任何东西。我不知道那个X是什么。但在自然界中，当我能写出这样的方程时，某些东西可以通过这个方程被建模： ($x^2 + 1 = 0$)。

数制的演变

👤 讲师：现在关键是，无论你谈到什么数制，我们都会看到某种数字。数字可以是正数。大家都同意吗？数字可以有多少种类型？正数。为什么？如果我们回到古代，也许一万年

Of course. Here is the processed content based on the provided subtitle text, following all specified requirements.

The Evolution of Number Systems

THE ORIGIN OF POSITIVE NUMBERS

 **Lecturer:** Now the point is, whatever the systems you talked about, so we will see some kind of numbers. The numbers could be positive. So, everyone agrees to it? How many types of numbers could be there? Positive numbers.

The lecturer then poses a series of philosophical questions to the class to explore the origin and necessity of numbers.

 **Lecturer:** Why someone, like we will go back to ages, maybe 10,000 years before, whenever this number system might have come up, why there will be a requirement of positive numbers? Why we have positive numbers? Are these tangible things? Numbers are tangible? It's a concept. Like the concept we are understanding, in understanding so many concepts we learn of and we assume. Do you think that numbers really exist? No, it's a concept. Like the language, do you think that language really exists? It's a concept. But through language, how many problems we solve? Likewise, what do you think, like why in mathematics, maybe ages before, positive numbers were thought of? Like we should have this kind of thing, positive number, something positive numbers. Why?

The lecturer then provides a concrete example to illustrate the need for positive numbers.

 **Lecturer:** Why, if I ask a problem: how many people are here in this room? So you need positive numbers to count for that. So for counting purpose... For somebody at back, whenever this thing might have come up, these people had a requirement of counting. Like we need to count. So just make all, I want you to assume yourself as a very primitive people, then you will understand this realization. Just assume that there is no mathematics, nothing at this time. Just understand that somebody comes up with an idea that, "See, we are finding it difficult how many of us are there. We should have some kind of mechanism to do it, a systematic way of... like we should have some kind of thing like 'how many' to just have a concept of 'how many'." This positive numbers come up. So all positive numbers for measurement. Now then you can extrapolate it to measure anything, any quantity, any metric. So you need positive numbers: one, two, three.

THE LIMITATIONS OF POSITIVE NUMBERS

The lecturer then shifts the discussion to the shortcomings of this initial number system.

 **Lecturer:** What's the problem with positive numbers? What sort of a problem we can't solve with positive numbers? What's the problem which you can't solve through positive numbers? Means not a problem. The point is: positive numbers allow us to model our counting problem. There was a counting problem, and the positive numbers are the one through which you can model it. So the problem was counting, and you have modeled it through positive numbers. Now, tell me what's the problem with the positive numbers?

Like what's the thing which you can't model through positive numbers? You can't simulate through positive numbers.

 **Student:** Huh?  **Student:** What?  **Student:** What is that which you can't do using positive numbers?  **Student:** Minus.  **Lecturer:** Okay, minus. Any other things? Do you think addition we can do using positive number? Yes. Addition we can do. So, suppose we have positive numbers 1 to infinity, till infinity. You take any two positive numbers...

数制的演变

正数的起源

 **讲师：**现在关键是，无论你谈到什么数制，我们都会看到某种数字。数字可以是正数。大家都同意吗？数字可以有多少种类型？正数。

讲师随后向课堂提出了一系列哲学性问题，以探讨数字的起源和必要性。

 **讲师：**为什么有人，比如我们回到古代，也许一万年，当这个数制可能出现的时候，为什么会有对正数的需求？为什么我们有正数？这些是具体的东西吗？数字是具体的吗？它是一个概念。就像我们理解的概念一样，在理解这么多概念的过程中，我们学习并假设它们。你认为数字真的存在吗？不，它是一个概念。就像语言，你认为语言真的存在吗？它是一个概念。但通过语言，我们解决了多少问题？同样地，你认为为什么在数学中，也许在远古时代，人们会想到正数？比如我们应该有这种东西，正数，某种正数。为什么？

讲师随后提供了一个具体的例子来说明正数的必要性。

 **讲师：**为什么，如果我提出一个问题：这个房间里有多少人？所以你需要正数来计数。所以为了计数目的.....对于后面的某个人来说，无论这件事何时发生，这些人都有计数的需求。比如我们需要计数。所以，我想让你把自己想象成一个非常原始的人，然后你就会理解这种认识。假设现在没有数学，什么都没有。只需要理解，有人提出了一个想法：“看，我们很难知道我们有多少人。我们应该有某种机制来做到这一点，一种系统化的方式.....比如我们应该有某种像‘多少’这样的东西，只是为了有一个‘多少’的概念。”于是正数就出现了。所以所有的正数都用于测量。然后你可以将其推广到测量任何东西、任何数量、任何度量。所以你需要正数：一、二、三。

正数的局限性

讲师随后将讨论转向了这个初始数制的缺点。

👨‍🏫 讲师：正数有什么问题？什么样的问题是我们用正数无法解决的？你无法通过正数解决的问题是什么？我的意思不是说它本身是个问题。关键是：正数让我们能够为我们的计数问题建模。有一个计数问题，而正数就是你可以用来建模它的工具。所以问题是计数，你已经通过正数对它进行了建模。现在，告诉我正数有什么问题？比如，有什么东西是你无法通过正数建模的？你无法通过正数模拟的。

👨‍🎓 学生：嗯？👨‍🎓 学生：什么？👨‍🎓 学生：有什么是使用正数做不到的？👨‍🎓 学生：减法。

👨‍🏫 讲师：好的，减法。还有其他东西吗？你认为加法我们可以用正数做吗？是的。加法我们可以做。所以，假设我们有从1到无穷大的正数。你取任意两个正数.....

好的，这是根据您的要求，对提供的字幕文本进行处理后的结果。

1. ZERO INFORMATION LOSS - ENGLISH ONLY

What is that which you can't do using positive numbers?

Minus.

Okay, minus.

Any other things?

Do you think addition we can do using positive number?

Yes.

Addition?

Addition we can do.

So, suppose we have positive numbers 1 to infinity till infinity.

You take any two positive numbers and the problem is...

You will get the answer from the same series.

Somewhere in positive numbers.

So addition problem we can solve.

Minus problem we can't solve.

So if you have suppose at that time there was a no concept of negative number.

So at that time somebody realized that okay 2.

If I will take 2 and subtract 5 out of it what will be the result?

5 minus 3 is 5 minus 2 is fine.

But 2 minus 5 what will be the result?

So negative, you can't solve using positive numbers.

So you need some another way of understanding that what that number will be.

So just imagine at that time, you have no concept of mathematics or nothing, and 2 minus 5 is one problem, and how you will represent it?

So to represent or to model it, negative numbers comes into picture.

Now, there is not only, okay, negative is one, like, fraction is one problem which you can't solve using positive numbers.

What is that spatial thing which you can't represent using positive numbers?

So if I have to say, there is no one in this row.

Nothing.

Emptiness.

If I have to represent emptiness in mathematics, 0 is that number.

So you have to bring zero with positive numbers.

What is that number series we call?

Natural numbers.

So zero.

Emptiness.

So in nature, if you have to represent emptiness, that means nothing is there, you need zero.

So natural number series is that number series.

Zero.

So that brings zero into it.

But still with zero and positive numbers, that's a natural number series.

It can't solve the problem which he mentioned, subtraction.

So subtraction means accounting problems.

So credit problem, debit problem, these kind of an accounting problem.

I have given you this much credit.

So that will be accounted as negative one in your account.

Okay.

So these kind of a problem, for that, what do you need to bring in?

Negative numbers.

Now, so that's your natural numbers.

So natural numbers also can solve some problems like subtraction.

Then comes your integers.

So integers can solve your addition, your subtraction.

So whatever you can, whatever the problem which could be modeled using addition and subtraction, you can solve them for integers.

Then why is there a requirement of rational numbers?

So let's assume that you have any integer 5, and if somebody will ask you, can you divide 5 in equal measures, can you divide 5 equally to 3 pe

2. STRUCTURAL REORGANIZATION - ENGLISH ONLY

The Limitations of Positive Numbers

The lecture begins by exploring the limitations of using only positive numbers to model problems.

- **Subtraction:** A core problem is subtraction. For example, $2 - 5$ cannot be solved within the set of positive numbers. This requires the concept of **negative numbers**.
- **Emptiness:** Representing the concept of "nothing" or "emptiness" is impossible with positive numbers alone. This requires the number **zero**.
- **Division:** The need for **rational numbers** arises from the inability to solve certain division problems with integers.

The Evolution of Number Systems

The lecture traces the historical development of number systems as a response to these limitations.

1. **Natural Numbers:** The series starting from zero (or one) and including all positive integers. This system can handle addition but fails at subtraction and representing emptiness.
2. **Integers:** By adding negative numbers to the natural numbers, we create the integers. This system can solve both addition and subtraction problems.
3. **Rational Numbers:** The next logical step is to address division. The lecture poses the question: can you divide the integer 5 equally among 3 people? This problem cannot be solved using integers alone, leading to the need for fractions or rational numbers.

3. LOGICAL HIGHLIGHTING - ENGLISH ONLY

The Limitations of Positive Numbers

The lecture begins by exploring the limitations of using only **positive numbers** to model problems.

- **Subtraction:** A core problem is subtraction. For example, $2 - 5$ cannot be solved within the set of positive numbers. This requires the concept of **negative numbers**.
- **Emptiness:** Representing the concept of "nothing" or "emptiness" is impossible with positive numbers alone. This requires the number **zero**.
- **Division:** The need for **rational numbers** arises from the inability to solve certain division problems with integers.

The Evolution of Number Systems

The lecture traces the historical development of number systems as a response to these limitations.

1. **Natural Numbers:** The series starting from zero (or one) and including all positive integers. This system can handle *addition* but fails at *subtraction* and representing *emptiness*.
2. **Integers:** By adding negative numbers to the natural numbers, we create the integers. This system can solve both *addition* and *subtraction* problems.
3. **Rational Numbers:** The next logical step is to address *division*. The lecture poses the question: can you divide the integer 5 equally among 3 people? This problem cannot be solved using integers alone, leading to the need for fractions or **rational numbers**.

4. INTERACTION EXTRACTION - ENGLISH ONLY

 **Student:** Minus.  **Lecturer:** Okay, minus. Any other things? Do you think addition we can do using positive number? Yes. Addition? Addition we can do.

5. FULL CHINESE TRANSLATION

1. 绝对忠实原则 - 中文

有什么是使用正数做不到的？

减法。

好的，减法。

还有其他东西吗？

你认为加法我们可以用正数做吗？

是的。

加法？

加法我们可以做。

所以，假设我们有从1到无穷大的正数。

你取任意两个正数，问题是.....

你会从同一个数列中得到答案。

就在正数中的某个位置。

所以加法问题我们可以解决。

减法问题我们无法解决。

所以，假设在那个时候，还没有负数的概念。

所以在那个时候，有人意识到，好吧，2。

如果我拿2减去5，结果会是什么？

5减3可以，5减2也可以。

但是2减5，结果会是什么？

所以是负数，你无法用正数解决。

所以你需要另一种方式来理解那个数字是什么。

所以想象一下，在那个时候，你没有任何数学概念或任何东西，2减5是一个问题，你要如何表示它？

所以为了表示或建模它，负数就出现了。

现在，不仅仅是负数，分数也是一个你无法用正数解决的问题。

那个你无法用正数表示的空间上的东西是什么？

所以如果我必须说，这一排没有人。

没有东西。

空无。

如果我必须在数学中表示空无，0就是那个数字。

所以你必须把零和正数放在一起。

我们叫那个数列什么？

自然数。

所以是零。

空无。

所以在自然界中，如果你必须表示空无，那意味着那里什么都没有，你需要零。

所以自然数数列就是那个数列。

零。

所以这就把零引入了进来。

但是，即使有了零和正数，那也只是自然数数列。

它仍然无法解决他提到的那个问题，减法。

所以减法意味着会计问题。

所以是贷方问题、借方问题，这类会计问题。

我给了你这么多信用额度。

那么在你的账户里，这将被记为负数。

好的。

所以对于这类问题，你需要引入什么？

负数。

现在，这就是你的自然数。

所以自然数也可以解决一些问题，比如减法。

然后是你的整数。

所以整数可以解决你的加法、你的减法。

所以任何你可以，任何可以用加法和减法建模的问题，你都可以用整数解决。

那么为什么还需要有理数呢？

所以让我们假设你有一个整数5，如果有人问你，你能把5分成相等的份数吗，你能把5平均分给3个人吗？

Of course. Here is the processed content based on the provided subtitle text, following all specified requirements.

1. Zero Information Loss & Structural Reorganization (English Only)

THE PROGRESSION OF NUMBER SYSTEMS: A FOUNDATIONAL ANALOGY

The lecturer uses a historical progression of number systems to illustrate a key point about mathematical expansion and problem-solving.

- **Integers** can solve problems modeled by addition and subtraction.
- **Rational numbers** (represented as m/n) are needed for problems like division. For example, dividing 5 bananas equally among 3 people cannot be solved with integers alone. The result, $5/3$, is a rational number.
- **Irrational numbers** exist in nature that cannot be represented as m/n . Examples include **Pi (π)** and $\sqrt{2}$. (Note: $22/7$ is only an approximation of π).

🧑 Student: Pi. 🧑 Lecturer: Pi, yes, pi.

The core point is that when existing number systems are insufficient to solve a problem, mathematics expands to create new ones.

THE CORE PROBLEM: $x^2 + 1 = 0$

The lecturer connects this historical progression to the current problem.

- We are now in a situation with the equation $x^2 + 1 = 0$.
- A vast array of existing numbers (positive, negative, rational, irrational) cannot solve this problem.
- However, just as we progressed from positive numbers to real numbers, we can progress further. Numbers are a concept, a tool to model nature and solve problems.
- Since existing numbers are insufficient, we must expand our number system.

🧑 Lecturer: Are you getting this kind of a progression which over a period of time happens? Are you grasping like where I am taking you? 🧑 Lecturer: Getting my point?

THE SOLUTION: INTRODUCING IMAGINARY NUMBERS

To solve $x^2 = -1$, we need a new kind of number.

- The operation $\sqrt{-1}$ is prohibited within the real number system.
- We assume the existence of an **imaginary number**.
- This special number is denoted by the symbol **iota (i)**.
- The fundamental assumption is: $\sqrt{-1} = i$.
- Based on this single assumption, the entire field of **complex algebra** and **complex mathematics** is built.

2. Full Chinese Translation (中文全文翻译)

数系的演进：一个基础类比

讲师使用数系的历史演进来说明关于数学扩展和问题解决的一个关键点。

- **整数** 可以解决由加法和减法建模的问题。

- **有理数**（表示为 m/n ）是解决除法等问题所必需的。例如，将5根香蕉平均分给3个人，仅用整数是无法解决的。结果 $5/3$ 是一个有理数。
- **无理数** 存在于自然界中，不能表示为 m/n 。例子包括 **圆周率 π** 和 $\sqrt{2}$ 。（注意： $22/7$ 只是 π 的近似值）。

👤 学生：圆周率。👤 讲师：圆周率，是的，圆周率。

核心观点是，当现有的数系不足以解决某个问题时，数学就会扩展以创造新的数系。

核心问题： $x^2 + 1 = 0$

讲师将这个历史演进与当前的问题联系起来。

- 我们现在面临方程 $x^2 + 1 = 0$ 的情况。
- 大量现有的数（正数、负数、有理数、无理数）都无法解决这个问题。
- 但是，正如我们从正数演进到实数一样，我们可以进一步演进。数是一个概念，是模拟自然和解决问题的工具。
- 由于现有的数系不足以解决问题，我们必须扩展我们的数系。

👤 讲师：你们明白这种随着时间的推移而发生的演进过程吗？你们理解我要把你们引向何方吗？👤 讲师：明白我的意思了吗？

解决方案：引入虚数

为了求解 $x^2 = -1$ ，我们需要一种新的数。

- 在实数系中，运算 $\sqrt{-1}$ 是被禁止的。
- 我们假设存在一个**虚数**。
- 这个特殊的数用符号 **iota (i)** 表示。
- 基本的假设是： $\sqrt{-1} = i$ 。
- 基于这一个假设，整个**复数代数**和**复数数学**的领域就建立起来了。

好的，这是根据您的要求，对提供的字幕文本进行处理后的结果。

THE NATURE OF ALPHA AND BETA

- In quantum computing, the state of a qubit is represented as $\alpha|0\rangle + \beta|1\rangle$.
- The coefficients α and β are not simple probabilities. They are **probability amplitudes**.

- The actual probability of measuring the state $|0\rangle$ is $|\alpha|^2$, and the probability of measuring $|1\rangle$ is $|\beta|^2$.
- Therefore, the fundamental rule is that $|\alpha|^2 + |\beta|^2 = 1$.
- Crucially, α and β can be, and most often are, **complex numbers**. This is why understanding complex numbers is essential for quantum computing.

 Lecturer: So that's why we have to understand what is alpha and beta? Probability. Not probabilities. Probability amplitudes. They are the amplitudes. So probability is square of it. Alpha square and beta square are the probabilities. So alpha square plus beta square should be equal to? 1. So these alpha and beta could be complex numbers.

THE IMAGINARY UNIT: i

To work with complex numbers, we first need to define the imaginary unit.

- The problem arises from the equation $x^2 = -1$, which has no solution in the real number system.
- Mathematicians solved this by making an assumption: let there be a number i such that $\sqrt{-1} = i$.
- This number i is called the **imaginary unit** or **iota**.
- From this single assumption, the entire field of **complex algebra** and **complex mathematics** is built.

 Lecturer: So in mathematics, the famous thing is let's assume. So let's assume square root of minus 1 is equal to i . And based on this assumption, the whole complex algebra, complex mathematics is based. And it helps us to model a lot of stuff.

KEY PROPERTIES OF i

Understanding the powers of i is fundamental.

- $i = \sqrt{-1}$
- $i^2 = -1$ (This is the most important property: the square of an imaginary number is a real number).
- $i^3 = i^2 * i = -i$
- $i^4 = (i^2)^2 = (-1)^2 = 1$

 Lecturer: So i is square root of minus 1. So what will be i square? Minus 1. i square is minus 1. So just this is the stuff which we need to know. Wherever i comes, i is nothing but square root of minus 1. And whenever you have a square root of i , then i is a imaginary

number, but i square will be? Your real number. So, i square will be minus 1 only. So, imaginary square is real. When you are doing imaginary twice, you are getting the real number. So, imaginary square is real number that is minus 1.

Calculating Higher Powers: i^{15}

 Student: What will be i raise to the power 15?  Lecturer: i raise to the power 15. i square is minus 1. 2 raise to the power 7... so that will be minus... so minus i . Right, yeah.

The calculation works as follows:

1. $i^{15} = i^{14} * i$
2. $i^{14} = (i^2)^7 = (-1)^7 = -1$
3. Therefore, $i^{15} = -1 * i = -i$

REPRESENTING COMPLEX NUMBERS

A complex number is a combination of a real part and an imaginary part.

- A complex number C is written in the form: $C = a + bi$
- a is the **real part**.
- b is the **imaginary part** (the coefficient of i).

 Lecturer: So this is the imaginary number which we have discussed, i . Now the complex number you will write it in this way. So complex number will be $a + bi$. What is a ? a is the real part. That's why the complex is complex is the combination of real and imaginary number. So real plus imaginary is complex. So C is a complex number and how... So now you will write C , a is the real part and b is the...

中文全文翻译

ALPHA 和 BETA 的性质

- 在量子计算中，一个量子比特的状态表示为 $\alpha|0\rangle + \beta|1\rangle$ 。
- 系数 α 和 β 不是简单的概率。它们是**概率幅**。
- 测量到状态 $|0\rangle$ 的实际概率是 $|\alpha|^2$ ，测量到 $|1\rangle$ 的概率是 $|\beta|^2$ 。
- 因此，基本规则是 $|\alpha|^2 + |\beta|^2 = 1$ 。

- 关键在于， α 和 β 可以是，而且大多数情况下就是**复数**。这就是为什么理解复数对于量子计算至关重要。

👤 讲师：所以这就是为什么我们必须理解 alpha 和 beta 是什么？概率。不是概率。是概率幅。它们是幅值。所以概率是它的平方。Alpha 平方和 beta 平方是概率。所以 alpha 平方加 beta 平方应该等于？1。所以这些 alpha 和 beta 可以是复数。

虚数单位： i

为了处理复数，我们首先需要定义虚数单位。

- 问题源于方程 $x^2 = -1$ ，它在实数系统中无解。
- 数学家通过做一个假设来解决这个问题：假设存在一个数 i ，使得 $\sqrt{-1} = i$ 。
- 这个数 i 被称为**虚数单位**或 **iota**。
- 基于这一个假设，整个**复数代数**和**复数数学**的领域就建立起来了。

👤 讲师：所以在数学中，著名的方法就是假设。所以我们假设根号下负一等于 i 。基于这个假设，整个复数代数、复数数学就建立起来了。它帮助我们模拟了很多东西。

i 的关键性质

理解 i 的幂次是基础。

- $i = \sqrt{-1}$
- $i^2 = -1$ （这是最重要的性质：虚数的平方是一个实数）。
- $i^3 = i^2 * i = -i$
- $i^4 = (i^2)^2 = (-1)^2 = 1$

👤 讲师：所以 i 是根号下负一。那么 i 的平方是什么？负一。 i 的平方是负一。所以这就是我们需要知道的东西。无论 i 出现在哪里， i 就是根号下负一。当你有一个 i 的平方根时，那么 i 是一个虚数，但 i 的平方会是？你的实数。所以， i 的平方只会是负一。所以，虚数的平方是实数。当你对虚数进行两次运算时，你得到的是实数。所以，虚数的平方是实数，即负一。

计算更高次幂： i^{15}

👤 学生： i 的 15 次方是多少？👤 讲师： i 的 15 次方。 i 的平方是负一。2 的 7 次方... 所以那将是负... 所以是负 i 。对，是的。

计算过程如下：

1. $i^{15} = i^{14} * i$
2. $i^{14} = (i^2)^7 = (-1)^7 = -1$
3. 因此, $i^{15} = -1 * i = -i$

复数的表示

一个复数是实数部分和虚数部分的组合。

- 一个复数 **C** 写成以下形式: $C = a + bi$
- **a** 是实部。
- **b** 是虚部 (**i** 的系数)。

 讲师: 这就是我们讨论过的虚数, i 。现在, 复数你会这样写。所以复数是 a 加 $b i$ 。 a 是什么? a 是实部。这就是为什么复数, 复数是实数和虚数的组合。所以实数加虚数就是复数。所以 C 是一个复数, 然后... 所以现在你会写 C , A 是实部, B 是...

Quantum Computing Lecture Notes (Continued)

Complex Numbers: Algebraic Interpretation

DEFINITION RECAP

- **Complex number:** $C = a + bi$
 - **a** is the **real part**
 - **b** is the **imaginary part** (coefficient of **i**)
- Real + Imaginary = Complex

 Lecturer: "So real plus imaginary is equal to complex. That's why it is complex. Real is also there in that number, imaginary is also there in that number."

TWO INTERPRETATIONS OF COMPLEX NUMBERS

This notation has different ways of understanding:

1. **Algebraic interpretation** – working with numbers through algebraic operations
2. **Graphical/Visual interpretation** – understanding complex numbers visually on a plane

👤 Lecturer: "First we will see the algebraic interpretation of the complex numbers. So how you will interpret any number which is written, how algebraically you will work it out."

👤 Lecturer: "The importance is when you will translate them into visualization. How visually you can understand them. So that's most important. You understand this algebraic part or not that's not so important, but understanding or sort of visualizing or that graphical understanding is important about complex numbers."

ALGEBRAIC OPERATIONS: ADDITION EXAMPLE

Given two complex numbers:

- $C1 = 5 + 3i$
- $C2 = -6 + 8i$

Addition rule: Real parts combine with real parts; imaginary parts combine with imaginary parts.

$$C1 + C2 = (5 + (-6)) + (3i + 8i) = -1 + 11i$$

👤 Lecturer: "How will you add two complex numbers? $C1$ plus $C2$: real will go with real, imaginary will go with imaginary. Simple enough. Minus six plus five is minus one, plus eight plus three is eleven. So, simple enough."

👤 Lecturer: "This addition is fine, it is more or less like any number system. If you have to add two natural numbers like 3 plus 2 is 5. Likewise, these two are the complex numbers."

GRAPHICAL VISUALIZATION OF COMPLEX NUMBERS

Complex numbers are visualized on a **two-dimensional plane**:

- **Real axis** (horizontal, like x-axis)
- **Imaginary axis** (vertical, like y-axis)

👤 Lecturer: "So this is my real axis and this is my imaginary axis. So this is, in your understanding, it is an x-axis, it is a y-axis."

Visualizing $5 + 3i$:

- Move 5 units along the **real axis** (positive direction)
- Move 3 units along the **imaginary axis** (positive direction)

👤 Lecturer: "In normal words, like if you have integers, so you have an integer line like this 0 1 2 3 4 because integer doesn't have the imaginary part so it is only on the real axis. So this is on the real axis only, so minus 1 minus 2 minus 3."

👤 Lecturer: "So here we will start from here and then we will go there. Just imagine that if you have to add 3 plus 2 and you have to visualize it. How will you visualize it? First you will draw 1, 2, 3..."

中文全文翻译

复数：代数解释

定义回顾

- 复数: $C = a + bi$
 - a 是实部
 - b 是虚部 (i 的系数)
- 实数 + 虚数 = 复数

👤 讲师: "所以实数加虚数等于复数。这就是为什么它叫复数。在那个数里, 既有实数也有虚数。"

复数的两种理解方式

这种记法有不同的理解方式:

1. 代数解释 – 通过代数运算来处理数字
2. 图形/视觉解释 – 在平面上直观地理解复数

👤 讲师: "首先我们来看复数的代数解释。你将如何解释任何写出来的数字, 以及如何用代数方法处理它。"

👤 讲师: "重要的是当你把它们转化为可视化时。你如何直观地理解它们。这才是最重要的。你是否理解代数部分并不那么重要, 但理解或可视化, 或者说对复数的图形理解才是重要的。"

代数运算：加法示例

给定两个复数：

- $C1 = 5 + 3i$
- $C2 = -6 + 8i$

加法规则： 实部与实部相加，虚部与虚部相加。

$$C1 + C2 = (5 + (-6)) + (3i + 8i) = -1 + 11i$$

👤 讲师："如何将两个复数相加？ $C1$ 加 $C2$ ：实部与实部相加，虚部与虚部相加。很简单。负六加五等于负一，加上八加三等于十一。所以，很简单。"

👤 讲师："这个加法没问题，它或多或少像任何数系一样。如果你要加两个自然数，比如 3 加 2 等于 5。同样地，这两个是复数。"

复数的图形可视化

复数在二维平面上可视化：

- **实轴**（水平方向，类似 x 轴）
- **虚轴**（垂直方向，类似 y 轴）

👤 讲师："这是我的实轴，这是我的虚轴。按照你的理解，这就像 x 轴和 y 轴。"

可视化 $5 + 3i$ ：

- 沿**实轴**正方向移动 5 个单位
- 沿**虚轴**正方向移动 3 个单位

👤 讲师："用通常的话说，就像整数一样，你有一条像这样的整数线：0 1 2 3 4，因为整数没有虚部，所以它只在实轴上。所以这只是在实轴上，还有负1 负2 负3。"

👤 讲师："所以我们从这里开始，然后走到那里。想象一下，如果你要加 3 加 2 并且要把它可视化。你会怎么可视化？首先你会画 1, 2, 3..."

Quantum Computing Lecture - Complex Number Visualization (Continued)

Complex Number Addition on the Complex Plane

VISUALIZING COMPLEX NUMBERS AS VECTORS

The lecturer continues the explanation of complex number representation using vector addition on the complex plane.

Key Concept: Complex numbers can be visualized as vectors on a 2D plane with:

- **Real axis** (horizontal)
- **Imaginary axis** (vertical)

EXAMPLE: REPRESENTING $5 + 3i$

 **Lecturer:** "Same concept I will apply here. 5 plus 3i. So what is the real part? 1, 2, 3, 4, 5."

 **Lecturer:** "What is the complex part?"

 **Student:** "3. 3i."

 **Lecturer:** "3i. And 3 is positive. So 1, 2, 3."

Visualization Process:

1. Walk **5 units** along the **real axis** (positive direction)
2. Walk **3 units** along the **imaginary axis** (positive direction)
3. The endpoint is the vector representing **$5 + 3i$**

 **Lecturer:** "Now the vector is? This is 5 plus 3i, this is the vector. So how I have done it: five units real and three units imaginary."

COMPARISON WITH INTEGER ADDITION

Integer Addition (Real Axis Only)	Complex Addition (Real + Imaginary Axes)
Walk along a single line	Walk along two perpendicular axes
$3 + 2 = 5$ units on real axis	5 units real + 3 units imaginary
One-dimensional movement	Two-dimensional movement

 **Lecturer:** "In mathematics, bases will always be same. So here the base is, you have to just walk that many units. The only difference is here we have just the real axis, and here we do have the imaginary axis."

EXAMPLE: REPRESENTING $-2 + 4i$

 **Lecturer:** "If you have to draw minus 6 plus 8i, minus 6. So this way, because I don't have enough space, so I will keep it small. So minus 2 plus maybe 4i."

Steps:

1. Locate **-2** on the real axis (2 units to the left of origin)
2. Move **4 units** upward along the imaginary axis
3. The endpoint represents **$-2 + 4i$**

 **Lecturer:** "So minus 2, 1, 2, and make it minus. So it goes. Or maybe plus 1. So 1, 2, 3, 4. So my vector is, this is minus 2 plus 4, right? Minus 2 plus 4i."

ADDING TWO COMPLEX VECTORS

 **Lecturer:** "These two vectors, you have to add them. So, addition means what? What we have done there for addition? 3 plus 2. What we have done? First, we traveled 3 unit. Then, wherever 3 unit was ending, from there, we started 2 units more. And then, we reached the 5th unit. So, 3 plus 2 is 5."

Vector Addition Principle:

- Start at the origin
- Travel along the first vector
- From the endpoint of the first vector, travel along the second vector
- The final position is the sum

中文全文翻译

量子计算课程 - 复数可视化（续）

复平面上的复数加法

将复数可视化为向量

讲师继续解释使用复平面上的向量加法来表示复数。

核心概念： 复数可以在二维平面上可视化为向量，包含：

- 实轴（水平方向）
- 虚轴（垂直方向）

示例：表示 $5 + 3i$

👤 讲师： "同样的概念我在这里应用。5 加 $3i$ 。那么实部是多少？ 1, 2, 3, 4, 5。"

👤 讲师： "虚部是多少？"

👤 学生： "3。 $3i$ 。"

👤 讲师： " $3i$ 。而且 3 是正的。所以 1, 2, 3。"

可视化过程：

1. 沿实轴走 5 个单位（正方向）
2. 沿虚轴走 3 个单位（正方向）
3. 终点就是表示 $5 + 3i$ 的向量

👤 讲师： "现在向量在哪里？ 这就是 5 加 $3i$ ，这就是向量。我是怎么做到的：实轴五个单位，虚轴三个单位。"

与整数加法的比较

整数加法（仅实轴）	复数加法（实轴 + 虚轴）
沿一条直线行走	沿两条垂直的轴行走
$3 + 2 =$ 实轴上 5 个单位	实轴 5 个单位 + 虚轴 3 个单位
一维运动	二维运动

👤 讲师： "在数学中，基底总是相同的。所以这里的基底是，你只需要走那么多单位。唯一的区别是这里我们只有实轴，而这里我们有虚轴。"

示例：表示 $-2 + 4i$

👤 讲师："如果你要画负 6 加 $8i$ ，负 6。那么这样，因为我没有足够的空间，所以我把它画小一点。所以负 2 加可能 $4i$ 。"

步骤：

1. 在实轴上找到 -2 （原点左侧 2 个单位）
2. 沿虚轴向上移动 4 个单位
3. 终点表示 $-2 + 4i$

👤 讲师："所以负 2，1，2，然后变成负的。就这样。或者可能加 1。所以 1，2，3，4。所以我的向量是，这是负 2 加 4，对吧？负 2 加 $4i$ 。"

两个复数向量的加法

👤 讲师："这两个向量，你要把它们加起来。那么，加法是什么意思？我们之前做加法是怎么做的？3 加 2。我们做了什么？首先，我们走了 3 个单位。然后，无论 3 个单位结束在哪里，从那里开始，我们又走了 2 个单位。然后，我们到达了第 5 个单位。所以，3 加 2 等于 5。"

向量加法原理：

- 从原点开始
- 沿第一个向量行走
- 从第一个向量的终点，沿第二个向量行走
- 最终位置就是它们的和

Of course. Here is the processed content based on the provided subtitle file, following all specified requirements.

1. Zero Information Loss & Structural Reorganization (English Only)

VECTOR ADDITION AND THE PARALLELOGRAM LAW

The concept of addition for vectors is analogous to addition on the real number line.

- **Real Number Addition (1D):** For $3 + 2$, you start at 0, travel 3 units, and then from that point, travel another 2 units. You arrive at 5.
- **Complex Number Addition (2D):** The same principle applies to vectors. To add $C1$ (e.g., $5 + 3i$) and $C2$ (e.g., $-2 + 4i$):
 1. Travel along the first vector $C1$ to its endpoint.
 2. From that endpoint, travel parallel to the second vector $C2$ for the same distance and direction.
 3. The final position reached is the sum.

This process forms a parallelogram, which is why it is called the **Parallelogram Law** of vector addition.

- **General Rule:** To add vector a and vector b , draw a parallelogram where a and b are adjacent sides. The diagonal of the parallelogram is the resultant vector $a + b$.

THE NUMBER SYSTEM: FROM 1D TO 2D AND BEYOND

- The real number system is **one-dimensional**, existing on a single line.
- The complex number system is **two-dimensional**, represented by the complex plane.
- This concept can be extended. In the future, if needed, a **three-dimensional number system** could be created.
 - A three-dimensional number system would be structured as: Real Number + (element for 2nd dimension) + (element for 3rd dimension).
 - The specific properties and uses of such a system are currently unknown, but the principle of extending the number line is valid.

SUBTRACTION OF COMPLEX NUMBERS

Subtraction is treated as the addition of a negative vector.

- **Concept of Negation:** On the real number line, -2 is a 180-degree turn from $+2$. The same applies to vectors. $-C2$ is vector $C2$ rotated by 180 degrees.
- **Procedure for $C1 - C2$:**
 1. Find the negative of $C2$ ($-C2$) by rotating it 180 degrees.
 2. Add $C1$ and $-C2$ using the standard vector addition method (Parallelogram Law).
- **Algebraic Example:**
 - If $C1 = 5 + 3i$ and $C2 = -2 + 4i$, then $C1 - C2$ is:
 - Real part: $5 - (-2) = 7$
 - Imaginary part: $3 - 4 = -1$

- Result: $7 - 1i$
- *Correction from transcript:* The lecturer said $3 - 4 = -8$, which is a verbal error. The correct calculation is $3 - 4 = -1$.

SUMMARY OF KEY CONCEPTS

- **Addition:** Travel along the first vector, then travel parallel to the second vector from the first's endpoint.
- **Parallelogram Law:** The diagonal of the parallelogram formed by two vectors is their sum.
- **Subtraction:** $C1 - C2$ is equivalent to $C1 + (-C2)$, where $-C2$ is $C2$ rotated by 180 degrees.

2. Logical Highlighting (English Only)

CORE CONCEPTS & TERMINOLOGY

- **Vector Addition:** The process of combining two vectors.
- **Parallelogram Law:** A geometric method for vector addition.
- **Complex Number System:** A **two-dimensional** number system.
- **One-Dimensional Number System:** The real number line.
- **Three-Dimensional Number System:** A hypothetical extension of the number system.
- **Negation of a Vector:** Rotating a vector by **180 degrees**.
- **Subtraction:** The addition of a negated vector.

MATHEMATICAL EXPRESSIONS

The algebraic subtraction of complex numbers is straightforward: $(a + bi) - (c + di) = (a - c) + (b - d)i$

3. Interaction Extraction (English Only)

No student interactions were detected in the provided transcript segment.

4. Full Chinese Translation (中文全文翻译)

向量加法与平行四边形法则

向量的加法概念与实数线上的加法类似。

- **实数加法 (一维):** 对于 $3 + 2$ ，你从 0 开始，移动 3 个单位，然后从那个点再移动 2 个单位。你到达了 5。
- **复数加法 (二维):** 同样的原理适用于向量。要计算 $C1$ (例如 $5 + 3i$) 和 $C2$ (例如 $-2 + 4i$) 的和：
 1. 沿着第一个向量 $C1$ 移动到它的终点。
 2. 从那个终点开始，平行于第二个向量 $C2$ 移动相同的距离和方向。
 3. 最终到达的位置就是它们的和。

这个过程形成了一个平行四边形，这就是为什么它被称为向量加法的**平行四边形法则**。

- **通用规则:** 要计算向量 a 和向量 b 的和，画一个以 a 和 b 为邻边的平行四边形。该平行四边形的对角线就是结果向量 $a + b$ 。

数系：从一维到二维及更高维度

- 实数系是**一维的**，存在于一条直线上。
- 复数系是**二维的**，由复平面表示。
- 这个概念可以扩展。将来如果需要，可以创建一个**三维数系**。
 - 一个三维数系的结构将是：实数 + (第二维的元素) + (第三维的元素)。
 - 这样一个系统的具体属性和用途目前未知，但扩展数轴的原则是有效的。

复数的减法

减法被视为加上一个负向量。

- **取反的概念:** 在实数线上， -2 是从 $+2$ 旋转 180 度得到的。这同样适用于向量。 $-C2$ 是向量 $C2$ 旋转 180 度后的结果。
- **$C1 - C2$ 的步骤:**
 1. 通过将 $C2$ 旋转 180 度来找到它的负数 ($-C2$)。
 2. 使用标准的向量加法方法 (平行四边形法则) 将 $C1$ 和 $-C2$ 相加。
- **代数示例:**
 - 如果 $C1 = 5 + 3i$ 且 $C2 = -2 + 4i$ ，那么 $C1 - C2$ 是：

- 实部: $5 - (-2) = 7$
- 虚部: $3 - 4 = -1$
- 结果: $7 - 1i$
- 来自字幕的更正: 讲师说 $3 - 4 = -8$, 这是一个口误。正确的计算是 $3 - 4 = -1$ 。

关键概念总结

- **加法:** 沿着第一个向量移动, 然后从第一个向量的终点平行于第二个向量移动。
- **平行四边形法则:** 由两个向量构成的平行四边形的对角线就是它们的和。
- **减法:** $C1 - C2$ 等价于 $C1 + (-C2)$, 其中 $-C2$ 是 $C2$ 旋转 180 度后的结果。

Structured Document: Quantum Computing Lecture (Continued)

Complex Number Operations (Continued)

SUBTRACTION VISUALIZATION

The lecturer continues explaining the visualization of complex number subtraction.

Algebraic Subtraction:

- For $C1 - C2$ where $C1 = 5 + 3i$ and $C2 = -2 + 4i$:
 - Real part: $5 - (-2) = 7$
 - Imaginary part: $3 - 4 = -1$

Visualization Principle:

- **Negative** in the number system corresponds to a **180-degree turn**
- Subtraction $C1 - C2$ is visualized as: take $C2$, rotate it 180 degrees (making it $-C2$), then perform addition
- The operation becomes simply an addition after the rotation

ASSIGNMENT: VISUALIZATION PROGRAMMING TASK

 **Lecturer:** "In next class, you have to do all these visualizations. So what you need to do, you have to ask user to enter two numbers, two complex numbers, and then first, both complex numbers, you have to visualize them."

Task Requirements:

- Plot the two complex numbers on a 2D plane (as demonstrated in class)
 - Implement visualization of operators:
 - **Addition operator**
 - **Subtraction operator**
 - Additional operators will be covered later
 - Ask the user to choose which operation visualization they want to see
-

MULTIPLICATION OF COMPLEX NUMBERS

 **Lecturer:** "What about multiplication? Multiplying two complex numbers, what will happen? How will you multiply two complex numbers. Have you done algebraic multiplication before? So same."

Algebraic Multiplication: Given two complex numbers:

- $C1 = A1 + B1i$
- $C2 = A2 + B2i$

The multiplication follows standard algebraic rules:

- Each term of $C1$ multiplies with every term of $C2$
- **Key rule:** Wherever i^2 appears, replace it with -1

$$\begin{aligned} C1 \times C2 &= (A1 + B1i) \times (A2 + B2i) \\ &= A1 \cdot A2 + A1 \cdot B2i + B1i \cdot A2 + B1i \cdot B2i \\ &= A1 \cdot A2 + (A1 \cdot B2 + B1 \cdot A2)i + B1 \cdot B2 \cdot i^2 \\ &= A1 \cdot A2 + (A1 \cdot B2 + B1 \cdot A2)i + B1 \cdot B2 \cdot (-1) \\ &= (A1 \cdot A2 - B1 \cdot B2) + (A1 \cdot B2 + B1 \cdot A2)i \end{aligned}$$

DIVISION OF COMPLEX NUMBERS

 **Lecturer:** "Now, division. If you have to divide C1 by C2, what will happen? C1 by C2. So A1 plus B1i divided by A2 plus B2i."

The Trick (Method):

To divide $C1 / C2$:

1. Multiply both numerator and denominator by the **complex conjugate** of the denominator
2. The complex conjugate of $A2 + B2i$ is $A2 - B2i$

$$\begin{aligned} C1 / C2 &= (A1 + B1i) / (A2 + B2i) \\ &= (A1 + B1i) \times (A2 - B2i) / (A2 + B2i) \times (A2 - B2i) \end{aligned}$$

Complex Conjugate Property:

- When a complex number is multiplied by its complex conjugate, the result is a **real number**
- $(A2 + B2i) \times (A2 - B2i) = A2^2 + B2^2$ (a real number)

 **Lecturer:** "So when you will multiply a number with its complex conjugate, you will get a real number. And there is a special significance of that real number. So this a square plus b square is not a random number."

Geometric Significance of $A^2 + B^2$:

 **Lecturer:** "Do you know what it is?"  **Student:** "I can't remember."

The lecturer explains using the complex plane representation:

- For a complex number $A1 + B1i$ plotted on the complex plane:
 - The horizontal distance is $A1$
 - The vertical distance is $B1$
 - These form a right triangle

The value $A^2 + B^2$ represents the square of the magnitude (length) of the complex number vector from the origin to the point (A, B) on the complex plane.

中文全文翻译

复数运算（续）

减法可视化

讲师继续解释复数减法的可视化。

代数减法：

- 对于 $C1 - C2$ ，其中 $C1 = 5 + 3i$ ， $C2 = -2 + 4i$ ：
 - 实部： $5 - (-2) = 7$
 - 虚部： $3 - 4 = -1$

可视化原理：

- 数系中的负数对应**180度旋转**
- 减法 $C1 - C2$ 的可视化方法是：将 $C2$ 旋转180度（得到 $-C2$ ），然后执行加法运算
- 旋转后，该操作就变成了简单的加法

作业：可视化编程任务

🦊 讲师：“下一节课，你们需要完成所有这些可视化。所以你们需要做的是，让用户输入两个数，两个复数，然后首先，将这两个复数都进行可视化。”

任务要求：

- 在二维平面上绘制两个复数（按照课堂演示的方式）
- 实现运算符的可视化：
 - 加法运算符
 - 减法运算符
 - 其他运算符将在后续课程中涉及
- 让用户选择他们想要查看哪种运算的可视化

复数的乘法

👤 讲师: "那么乘法呢? 两个复数相乘, 会发生什么? 你们之前做过代数乘法吗? 是一样的。"

代数乘法: 给定两个复数:

- $C1 = A1 + B1i$
- $C2 = A2 + B2i$

乘法遵循标准代数规则:

- $C1$ 的每一项与 $C2$ 的每一项相乘
- 关键规则: 凡是出现 i^2 的地方, 都替换为 -1

$$\begin{aligned} C1 \times C2 &= (A1 + B1i) \times (A2 + B2i) \\ &= A1 \cdot A2 + A1 \cdot B2i + B1i \cdot A2 + B1i \cdot B2i \\ &= A1 \cdot A2 + (A1 \cdot B2 + B1 \cdot A2)i + B1 \cdot B2 \cdot i^2 \\ &= A1 \cdot A2 + (A1 \cdot B2 + B1 \cdot A2)i + B1 \cdot B2 \cdot (-1) \\ &= (A1 \cdot A2 - B1 \cdot B2) + (A1 \cdot B2 + B1 \cdot A2)i \end{aligned}$$

复数的除法

👤 讲师: "现在讲除法。如果要用 $C1$ 除以 $C2$, 会怎样? $C1$ 除以 $C2$ 。也就是 $A1$ 加 $B1i$ 除以 $A2$ 加 $B2i$ 。"

技巧 (方法):

要计算 $C1 / C2$:

1. 将分子和分母同时乘以分母的共轭复数
2. $A2 + B2i$ 的共轭复数是 $A2 - B2i$

$$\begin{aligned} C1 / C2 &= (A1 + B1i) / (A2 + B2i) \\ &= (A1 + B1i) \times (A2 - B2i) / (A2 + B2i) \times (A2 - B2i) \end{aligned}$$

共轭复数的性质:

- 当一个复数与其共轭复数相乘时, 结果是一个实数

- $(A^2 + B^2i) \times (A^2 - B^2i) = A^2^2 + B^2^2$ (一个实数)

👤 讲师: "所以当你用一个数乘以它的共轭复数时, 你会得到一个实数。而这个实数有一个特殊的意义。所以这个 $a^2 + b^2$ 并不是一个随机的数字。"

$A^2 + B^2$ 的几何意义:

👤 讲师: "你们知道它是什么吗?" 🧑 学生: "我记不起来了。"

讲师使用复平面表示进行解释:

- 对于在复平面上绘制的复数 $A_1 + B_1i$:
 - 水平距离是 A_1
 - 垂直距离是 B_1
 - 它们构成一个直角三角形

$A^2 + B^2$ 的值表示从原点到复平面上点 (A, B) 的复数向量的模长 (长度) 的平方。

Quantum Computing Lecture - Complex Number Operations (Continued)

1. Visualizing Complex Number Multiplication

THE LENGTH OF A COMPLEX VECTOR

👤 Lecturer: "Do you know what it is?" 🧑 Student: "I can't remember." 🧑 Lecturer: "What it is?"

The lecturer continues explaining the geometric interpretation of complex numbers:

- For a complex number $a_1 + b_1i$ represented on the complex plane:
 - The horizontal distance is a_1
 - The vertical distance is b_1
 - These form a right triangle

👤 Lecturer: "What is the length of this triangle?" 🧑 Student: "What is the length of a_1 ? What will be the length of this?" 🧑 Student: "It's root over $a^2 + b^2$." 🧑 Lecturer: " $a^2 + b^2$? Root over the whole."

Key Insight: The value $a^2 + b^2$ represents the **square of the length** of the complex vector from the origin to point (a, b) on the complex plane.

 **Lecturer:** "So, that's what this special number is. It is the square of the length of that vector. If you multiply a complex number with its complex conjugate, you get the square of the length of that vector: a squared plus b squared."

COMPLEX DIVISION

The lecturer explains the division operation for complex numbers:

- $c1 / c2 = (a1 + b1i) / (a2 + b2i)$
- Multiply numerator and denominator by the conjugate of the denominator:
 - $(a1 + b1i) * (a2 - b2i) / (a2 + b2i) * (a2 - b2i)$
- The denominator becomes $a2^2 + b2^2$, which is a **real number**
- The result will have both a **real part** and an **imaginary part**

 **Lecturer:** "Whatever this number comes, you have to just divide those numbers. So it will have a real part, and it will have an imaginary part. But always when you will divide, it will be $a2^2 + b2^2$, which is a real number."

VISUALIZING MULTIPLICATION ON THE COMPLEX PLANE

 **Lecturer:** "Addition and subtraction were easy to visualize using parallelogram law. So we were able to easily visualize what is happening when you are adding two complex numbers, what is happening when you are subtracting two complex numbers. Can you visualize what will happen when you will multiply two complex numbers? Can you visualize on that complex plane that when two vectors are getting multiplied, what is happening?"

Example: Multiplying 3 and i

The lecturer provides a concrete example:

- **Vector 1 ($a1$):** $3 + 0i$ (a real vector on the horizontal axis)
- **Vector 2 ($a2$):** $0 + i$ (a purely imaginary vector on the vertical axis)

 **Lecturer:** "So when you are multiplying these two, what will be the result? Simple enough? i and 3. So what will be the result? $3i$. Somewhere here, $3i$. This is my result."

Geometric Interpretation:

- Starting with 3 (on the real axis) and multiplying by i
- The result is $3i$ (on the imaginary axis)
- The vector has **rotated** from the real axis to the imaginary axis

 **Lecturer:** "So what kind of interpretation can you do? 3 and i . What happened? $3i$. $3i$, but visually now it is here. So what happened now? It went to the i ..."

量子计算课程 - 复数运算（续）

1. 复数乘法的可视化

复数向量的长度

 讲师: "你们知道它是什么吗?"  学生: "我记不起来了。"  讲师: "它是什么?"

讲师继续解释复数的几何意义:

- 对于在复平面上表示的复数 $a1 + b1i$:
 - 水平距离是 $a1$
 - 垂直距离是 $b1$
 - 它们构成一个直角三角形

 讲师: "这个三角形的长度是多少?"  讲师: " $a1$ 的长度是多少? 这个的长度是多少?"  学生: "是根号下 a 平方加 b 平方。"  讲师: " a 平方加 b 平方? 整个开根号。"

关键洞察: 值 $a^2 + b^2$ 表示从原点到复平面上点 (a, b) 的复数向量的长度的平方。

 讲师: "所以, 这就是这个特殊数字的含义。它是那个向量长度的平方。如果你将一个复数与其共轭复数相乘, 你会得到该向量长度的平方: a 平方加 b 平方。"

复数除法

讲师解释复数的除法运算:

- $c1 / c2 = (a1 + b1i) / (a2 + b2i)$
- 将分子和分母同时乘以分母的共轭复数：
 - $(a1 + b1i) * (a2 - b2i) / (a2 + b2i) * (a2 - b2i)$
- 分母变为 $a2^2 + b2^2$ ，这是一个实数
- 结果将同时包含实部和虚部

👤 讲师："无论得到什么数字，你只需对这些数字进行除法运算。所以它会有一个实部，也会有一个虚部。但当你做除法时，分母总是 $a2$ 平方加 $b2$ 平方，这是一个实数。"

在复平面上可视化乘法

👤 讲师："加法和减法很容易用平行四边形法则来可视化。所以我们能够很容易地看到两个复数相加时发生了什么，两个复数相减时发生了什么。你们能想象两个复数相乘时会发生什么吗？你们能在复平面上看到两个向量相乘时发生了什么吗？"

示例：3 乘以 i

讲师提供了一个具体示例：

- 向量 1 ($a1$): $3 + 0i$ (水平轴上的实数向量)
- 向量 2 ($a2$): $0 + i$ (垂直轴上的纯虚数向量)

👤 讲师："那么当这两个相乘时，结果会是什么？很简单吧？ i 和 3。结果是什么？ $3i$ 。大概在这里， $3i$ 。这就是我的结果。"

几何解释：

- 从 3 (在实轴上) 开始，乘以 i
- 结果是 $3i$ (在虚轴上)
- 向量从实轴旋转到了虚轴

👤 讲师："那么你能做什么样的解释呢？3 和 i 。发生了什么？ $3i$ 。 $3i$ ，但现在在视觉上它在这里。那么发生了什么？它到了 i 上....."

Lecture Content: Complex Number Multiplication & Visualization

1. Multiplication by i – Geometric Interpretation

CORE CONCEPT: PHASE ROTATION

When any vector (real number) is multiplied by the imaginary unit i , the result is a **90-degree phase rotation** while preserving the original length.

Example Walkthrough:

- Starting value: 3 (on the real axis)
- Multiply by i : $3 \times i = 3i$
- Result: $3i$ (on the imaginary axis)
- The vector has rotated **90 degrees counterclockwise**

 **Lecturer:** "So what kind of interpretation can you do? 3 and i . What happened? $3i$. $3i$, but visually now it is here. So what happened now? It went to the i part."

KEY PROPERTIES OF i MULTIPLICATION:

- **Length preservation:** The magnitude remains unchanged
 - $|3| = 3$
 - $|3i| = 3$
- **Phase introduction:** Exactly **90 degrees** phase difference
- **Repeated application:** Each multiplication by i adds another 90° rotation

Sequential Rotations (90° each step):

1. $3 \rightarrow$ multiply by $i \rightarrow 3i$ (90° rotation)
2. $3i \rightarrow$ multiply by $i \rightarrow 3i^2 = -3$ (180° rotation)
3. $-3 \rightarrow$ multiply by $i \rightarrow -3i$ (270° rotation)
4. $-3i \rightarrow$ multiply by $i \rightarrow 3$ (360° rotation, back to start)

 **Student:** "3."  **Lecturer:** "It will go back to there."

GENERAL PRINCIPLE:

 **Lecturer:** "So the point is multiplying any vector with an i will introduce a phase difference of 90 degree. It will keep the length of that vector same but it will just start flipping it 90, 90, 90."

2. General Complex Number Multiplication

PHASE DIFFERENCE BEHAVIOR

- **i is a special case:** Always introduces exactly 90° phase difference
- **General vectors:** Any random vector will introduce some *random* phase difference
- **Key insight:** When two vectors multiply, there will always be some phase difference change

 **Lecturer:** "Any random vector will introduce some random phase difference. Getting my point? So then it will be tricky like what will be the phase difference. But the point is when two vectors get multiplied, there will be some kind of phase difference change."

WHY I IS SPECIAL:

 **Lecturer:** "Here it is easy to figure out because I will always introduce the phase difference of 90 degrees. That's what I is so important."

3. Advanced Operations (Preview)

DIVISION AND SQUARE ROOTS

The lecturer notes that division and square root operations on complex numbers are more complex and will be discussed later:

 **Lecturer:** "Division, how to interpret division, how to interpret square root of anything that is kind of bit complex so when you will understand a lot of other stuff then we will discuss about those visualization that when you will divide two complex number how you will interpret and visualize it."

WHY VISUALIZATION MATTERS

 **Lecturer:** "Visualization is important for you because we are not going to solve it any mathematical equations or mathematical expressions concerning algebra using complex numbers. But you should be able to visualize that with these vectors, what is happening."

4. Connection to Quantum Computing

STATE VECTORS AND COMPLEX AMPLITUDES

The lecturer establishes the critical link between complex numbers and quantum state vectors:

 **Lecturer:** "Now, all these vectors like A plus B is something which I am showing you as a complex number. They will come as an amplitude of those state vectors. So when I will write a state vector, the amplitude of those state will be these complex numbers."

Key Takeaway:

- Complex numbers form the **amplitudes** of quantum state vectors
- Understanding complex number visualization is **essential** for quantum computing

 **Lecturer:** "So that's why understanding complex numbers and particularly visualization is important. You can be able to visualize it."

中文全文翻译

1. 乘以 i – 几何解释

核心概念：相位旋转

当任何向量（实数）乘以虚数单位 i 时，结果会产生 **90 度的相位旋转**，同时保持原始长度不变。

示例演示：

- 起始值：3（在实轴上）
- 乘以 i ： $3 \times i = 3i$
- 结果： $3i$ （在虚轴上）
- 向量旋转了 **90 度（逆时针）**

 **讲师：** "那么你能做什么样的解释呢？3 和 i 。发生了什么？ $3i$ 。 $3i$ ，但现在在视觉上它在这里。那么发生了什么？它到了 i 部分上。"

I 乘法的关键性质：

- 长度保持不变：幅度不变

- $|3| = 3$
- $|3i| = 3$

- 引入相位：恰好 **90 度** 的相位差
- 重复应用：每次乘以 i 都会增加 90° 旋转

连续旋转（每一步 90° ）：

1. $3 \rightarrow$ 乘以 $i \rightarrow 3i$ (90° 旋转)
2. $3i \rightarrow$ 乘以 $i \rightarrow 3i^2 = -3$ (180° 旋转)
3. $-3 \rightarrow$ 乘以 $i \rightarrow -3i$ (270° 旋转)
4. $-3i \rightarrow$ 乘以 $i \rightarrow 3$ (360° 旋转，回到起点)

🧑 学生："3。" 🧑 讲师："它会回到那里。"

一般原则：

🧑 讲师："所以关键在于，任何向量乘以 i 都会引入 90 度的相位差。它会保持该向量的长度不变，但会开始以 90 、 90 、 90 度翻转它。"

2. 一般复数乘法

相位差行为

- i 是一个特例：总是恰好引入 90° 相位差
- 一般向量：任何随机向量都会引入一些*随机*的相位差
- 关键洞察：当两个向量相乘时，总会发生某种相位差变化

🧑 讲师："任何随机向量都会引入一些随机的相位差。明白我的意思吗？那么问题就会变得棘手，比如相位差会是多少。但关键在于，当两个向量相乘时，总会发生某种相位差变化。"

为什么 i 很特殊：

🧑 讲师："这里很容易计算出来，因为 i 总是引入 90 度的相位差。这就是 i 如此重要的原因。"

3. 高级运算（预告）

除法和平方根

讲师指出，复数的除法和平方根运算更为复杂，将在后续讨论：

👤 讲师："除法，如何解释除法，如何解释任何东西的平方根，这有点复杂。所以当你理解了很多其他内容后，我们会讨论这些可视化问题，即当你将两个复数相除时，如何解释和可视化它。"

为什么可视化很重要

👤 讲师："可视化对你来说很重要，因为我们不会去解任何涉及复数代数的数学方程或数学表达式。但你应该能够通过这些向量可视化正在发生的事情。"

4. 与量子计算的关联

态向量和复数振幅

讲师建立了复数与量子态向量之间的关键联系：

👤 讲师："现在，所有这些像 A 加 B 这样的向量，就是我展示给你们的复数。它们将成为那些态向量的振幅。所以当我写一个态向量时，那些态的振幅就是这些复数。"

关键点：

- 复数构成了量子态向量的振幅
- 理解复数可视化对于量子计算至关重要

👤 讲师："所以这就是为什么理解复数，特别是可视化，很重要。你应该能够可视化它。"

好的，作为您的严谨学术助理，我已根据您提供的字幕文件和所有处理要求，完成了对后续文本的处理。

1. ZERO INFORMATION LOSS & STRUCTURAL REORGANIZATION (ENGLISH ONLY)

Complex Numbers as Amplitudes

The lecturer reiterates the fundamental role of complex numbers in quantum computing. The complex numbers visualized earlier are not just abstract concepts; they directly form the **amplitudes** of quantum **state vectors**.

 **Lecturer:** “They will come as an amplitude of those state vectors. So when I will write a state vector, the amplitude of those state will be these complex numbers.”

This reinforces why the visualization of complex numbers is **critical** for the course.

 **Lecturer:** “So that's why understanding complex numbers and particularly visualization is important. You can be able to visualize it.”

Representing Complex Numbers: Cartesian vs. Polar Coordinates

The lecturer moves from the geometric visualization to formal mathematical representations, which are essential for quantum computing.

1. Cartesian (Rectangular) Representation: A complex number like $5 + 3i$ can be represented as an ordered pair $(5, 3)$.

- The first position holds the **real part** (5).
- The second position holds the **imaginary part** (3).
- The imaginary unit i is not stored; the pair $(5, 3)$ is sufficient for representation.

 **Lecturer:** “So one way of representing 5 plus 3i is I can represent it like 5 and 3... I need an array having two positions... first position I will like the real part and the next part I will like the imaginary part... so 5 and 3 is enough for me to represent any complex number.”

2. Polar Representation: The lecturer transitions to polar coordinates, linking back to the **Bloch sphere** concept previously discussed. To uniquely locate a point like $5 + 3i$ in polar form, two pieces of information are needed:

- **Angle (θ):** The deviation from the positive x-axis.
- **Length (r):** The magnitude or length of the vector from the origin to the point.

 **Lecturer:** “So I need this angle... So here we are representing it as theta... then what I need to reach out to its edge it's the length... So this is the length of the vector.”

The lecturer generalizes this concept to higher dimensions:

- **2D:** 1 angle and 1 length.
- **3D:** 2 angles and 1 length.
- **4D:** 3 angles and 1 length.

 **Lecturer:** “If I will go to the three-dimension, then what I need? Two angles and one length. I will go to the four-dimension, three angles and one length.”

2. LOGICAL HIGHLIGHTING (ENGLISH ONLY)

Complex Numbers as Amplitudes

- The lecturer reiterates the fundamental role of complex numbers in quantum computing. The complex numbers visualized earlier are not just abstract concepts; they directly form the **amplitudes** of quantum **state vectors**.
- This reinforces why the visualization of complex numbers is **critical** for the course.

Representing Complex Numbers: Cartesian vs. Polar Coordinates

- **1. Cartesian (Rectangular) Representation:**
 - A complex number like $5 + 3i$ can be represented as an ordered pair $(5, 3)$.
 - The first position holds the **real part** (5).
 - The second position holds the **imaginary part** (3).
 - The imaginary unit i is not stored; the pair $(5, 3)$ is sufficient for representation.
 - **2. Polar Representation:**
 - The lecturer transitions to polar coordinates, linking back to the **Bloch sphere** concept previously discussed.
 - To uniquely locate a point like $5 + 3i$ in polar form, two pieces of information are needed:
 - **Angle (θ):** The deviation from the positive x-axis.
 - **Length (r):** The magnitude or length of the vector from the origin to the point.
 - The lecturer generalizes this concept to higher dimensions:
 - **2D:** 1 angle and 1 length.
 - **3D:** 2 angles and 1 length.
 - **4D:** 3 angles and 1 length.
-

3. INTERACTION EXTRACTION (ENGLISH ONLY)

 **Lecturer:** “Now, can you tell me in terms of a polar coordinate, if I have a. What is something essential for me to notice?”

 **Student:** “I know.”

 **Lecturer:** “Huh? What's-- here. You're right. So tell us again.”

 **Student:** “Huh?”

 **Lecturer:** “So I need, I am on center, starting point, and then all these four directions are going. So I need this angle.”

4. FULL CHINESE TRANSLATION (中文全文翻译)

复数作为振幅

讲师重申了复数在量子计算中的基本作用。之前可视化的复数不仅仅是抽象概念；它们直接构成了量子态向量的**振幅**。

 **讲师:** “它们将成为那些态向量的振幅。所以当我写一个态向量时，那些态的振幅就是这些复数。”

这再次强调了复数可视化对于本课程是**至关重要的**。

 **讲师:** “所以这就是为什么理解复数，特别是可视化，很重要。你应该能够可视化它。”

表示复数：笛卡尔坐标 vs. 极坐标

讲师从几何可视化转向了正式的数学表示，这对于量子计算至关重要。

1. 笛卡尔（直角坐标）表示法: 像 $5 + 3i$ 这样的复数可以表示为一个有序对 $(5, 3)$ 。

- 第一个位置存放**实部** (5)。
- 第二个位置存放**虚部** (3)。
- 虚数单位 i 不需要存储；对 $(5, 3)$ 就足以表示任何复数。

👤 讲师：“所以表示 5 加 $3i$ 的一种方法是，我可以把它表示为 5 和 3.....我需要有一个有两个位置的数组.....第一个位置我放实部，下一个位置我放虚部.....所以 5 和 3 对我来说就足以表示任何复数了。”

2. 极坐标表示法： 讲师过渡到极坐标，并将其与之前讨论过的**布洛赫球**概念联系起来。为了在极坐标形式下唯一确定像 $5 + 3i$ 这样的点，需要两条信息：

- **角度 (θ)：** 与正 x 轴的偏离角度。
- **长度 (r)：** 从原点到该点的向量的大小或长度。

👤 讲师：“所以我需要这个角度.....这里我们用 θ 来表示它.....然后我需要知道什么才能到达它的边缘，那就是它的长度.....所以这是向量的长度。”

讲师将这个概念推广到更高维度：

- **2D：** 1 个角度和 1 个长度。
- **3D：** 2 个角度和 1 个长度。
- **4D：** 3 个角度和 1 个长度。

👤 讲师：“如果我去到三维，那么我需要什么？两个角度和一个长度。我去到四维，三个角度和一个长度。”

师生互动

👤 讲师：“现在，你能告诉我，用极坐标来说，如果我有一个.....我需要注意到什么基本的东西吗？”

👤 学生：“我知道。”

👤 讲师：“嗯？什么——这里。你说得对。那你再说一遍。”

👤 学生：“嗯？”

👤 讲师：“所以我需要，我在中心，起点，然后所有这四个方向都在延伸。所以我需要这个角度。”

Quantum Computing Lecture - Polar Coordinates & Complex Numbers

1. Vector Representation in Polar Coordinates

2D VECTOR COMPONENTS

- **Length (magnitude):** This is the length of the vector
- **Angle (θ):** A deviation with respect to the positive x-axis

 **Lecturer:** "So a deviation with respect to positive x-axis, that's the theta. So these two things I need."

DIMENSIONAL SCALING

The number of angles required scales with dimensionality:

- **2D:** One angle is sufficient
- **3D:** Two angles are necessary
- **4D:** Three angles are necessary

 **Lecturer:** "If I will go to the three-dimension, then what I need? Two angles and one length. I will go to the four-dimension, three angles and one length. That's what we discussed last time."

APPLICATION TO COMPLEX NUMBERS

For a complex number $5 + 3i$:

- Need the **length** of the vector
- Need the **angle**

 **Lecturer:** "So same way, 5 plus 3i, you need a length of a vector, and you need angle. This is much more important for us."

2. Mathematical Conversion Formulas

FINDING THE LENGTH (MAGNITUDE)

$$\text{Length} = \sqrt{a^2 + b^2}$$

FINDING THE ANGLE (θ)

$$\theta = \tan^{-1}(b/a)$$

 **Lecturer:** "So if I have to ask you, tan theta is?"

 **Student:** "It's a trigonometry."

 **Lecturer:** "Yes. So trigonometry means, tan theta means perpendicular by base. So this is base and this is perpendicular. So B by A. Theta will be then tan inverse of B by A."

CONVERSION BACK FROM POLAR TO CARTESIAN

$$A = \rho \cos \theta$$

$$B = \rho \sin \theta$$

 **Lecturer:** "Can you understand how if you have the polar coordinate and you have to go back to its Cartesian? A will be rho cos theta and B will be rho sin theta."

3. Script Requirements

FOUR TASKS TO IMPLEMENT IN YOUR SCRIPT

1. **Visualization of a vector**
2. **Addition and subtraction** of vectors, followed by visualization
3. **Multiplication** of vectors (Note: Division is tricky - do not implement division)
4. **Conversion from Cartesian coordinates to polar coordinates**

 **Lecturer:** "So how many things you have to do? First, visualization of a vector. Second, addition and subtraction and then its visualization. Third, division and multiplication. No, no, division. Don't do division. That's a little bit tricky. Next is this one. Converting Cartesian coordinate into polar coordinates. So these four things you have to do in your script."

ADDITIONAL NOTES

- The conversion back from polar to Cartesian is provided in the program
- Understanding this reverse conversion is optional but beneficial

 **Lecturer:** "It's not so important, but if you can understand, that's fine. If you can, that is also fine."

4. Core Concept: Complex Numbers as Polar Representations

 **Lecturer:** "So now can you understand this statement related with polar coordinates related with how you are representing any complex number in polar coordinates? You are writing with its length and you are having one angle there. So a complex number is nothing but it is representing two stuff together: one is magnitude and another is space. So in nature when we need a single number to represent two quantities, so two matrices I need to represent using a single number."

中文全文翻译

1. 极坐标中的向量表示

二维向量分量

- **长度 (模长):** 这是向量的长度
- **角度 (θ):** 相对于正 x 轴的偏差

 **讲师:** "所以相对于正 x 轴的偏差, 那就是 θ 。所以我需要这两个东西。"

维度缩放

所需角度的数量随维度变化:

- **二维:** 一个角度就足够了
- **三维:** 需要两个角度

- 四维：需要三个角度

👤 讲师： "如果我要进入三维，那么我需要什么？两个角度和一个长度。如果我要进入四维，三个角度和一个长度。这就是我们上次讨论的内容。"

在复数中的应用

对于复数 $5 + 3i$ ：

- 需要向量的长度
- 需要角度

👤 讲师： "同样地，5 加 3i，你需要一个向量的长度，还需要角度。这对我们来说重要得多。"

2. 数学转换公式

求长度（模长）

$$\text{长度} = \sqrt{a^2 + b^2}$$

求角度 (θ)

$$\theta = \tan^{-1}(b/a)$$

👤 讲师： "如果我要问你， $\tan\theta$ 是什么？"

👤 学生： "这是三角函数。"

👤 讲师： "是的。所以三角函数意味着， $\tan\theta$ 等于对边除以邻边。所以这是底边，这是对边。所以是B除以A。那么 θ 就是 $\tan^{-1}(B/A)$ 。"

从极坐标转换回笛卡尔坐标

$$\begin{aligned} A &= \rho \cos \theta \\ B &= \rho \sin \theta \end{aligned}$$

👤 讲师: "你能理解如果你有极坐标, 如何转换回笛卡尔坐标吗? A将是 $\rho \cos \theta$, B将是 $\rho \sin \theta$ 。"

3. 脚本要求

脚本中需要实现的四个任务

1. 向量的可视化
2. 向量的加法和减法, 然后进行可视化
3. 向量的乘法 (注意: 除法很棘手——不要实现除法)
4. 从笛卡尔坐标到极坐标的转换

👤 讲师: "所以你需要做多少件事? 第一, 向量的可视化。第二, 加法和减法及其可视化。第三, 除法和乘法。不, 不, 除法。不要做除法。那有点棘手。接下来是这个。将笛卡尔坐标转换为极坐标。所以这四件事你需要在你的脚本中完成。"

附加说明

- 从极坐标转换回笛卡尔坐标的程序已经提供
- 理解这种反向转换是可选的, 但有益处

👤 讲师: "这并不那么重要, 但如果你能理解, 那很好。如果你能, 那也很好。"

4. 核心概念: 复数作为极坐标表示

👤 讲师: "所以现在你能理解这个关于极坐标的陈述吗? 关于你如何用极坐标表示任何复数? 你是用它的长度来写的, 并且那里有一个角度。所以一个复数本质上就是同时表示两个东西: 一个是模长, 另一个是空间。所以在自然界中, 当我们需要一个单一数字来表示两个量时, 我需要用单一数字来表示两个矩阵。"

5. Core Concepts: Complex Numbers as Probabilistic Amplitudes

5.1 The Dual Role of Complex Numbers

 **Lecturer:** "So now you understand the statement about polar coordinates? About how you represent any complex number in polar coordinates? You are writing it with its length, and you have one angle there. So a complex number is nothing but it is representing two things together: one is magnitude, and another is space."

In nature, when we need a single number to represent two quantities—for example, when I need to represent two matrices using a single number—I will use the complex number. This is because it gives me both:

- **The magnitude**
- **The phase difference**

REAL-WORLD ANALOGY: ELECTRICAL SIGNALS

 **Lecturer:** "In electrical signals, you need a magnitude and a phase difference, so complex numbers are heavily used there. The same way here, in our quantum computing case, I need to represent—for those states—there will be a magnitude associated with it, and there will be a phase difference as well."

5.2 Why Complex Numbers for Quantum Computing

 **Lecturer:** "That's why my submission before you is: that's why we are understanding complex numbers and we are using them heavily to represent our **probabilistic amplitude**. Because that probabilistic amplitude will not only reveal to me the magnitude—which is directly related to probability—it will also give me and indicate to me what is the phase difference. Getting my point? What is the phase difference? Not understanding? Okay."

5.3 The Qubit State Representation

So this is my $\alpha|1\rangle + \beta|0\rangle$. What are the primary states? What are the primary states which make up a qubit?

A **qubit** is a combination of the $|1\rangle$ and $|0\rangle$ vectors. These are the two vectors which somehow make up the qubit. And what is that "somehow"?

- α and β are the **probabilistic magnitudes**
- α and β are the **probabilistic amplitudes**

EXAMPLE: REAL-NUMBER PROBABILISTIC AMPLITUDES

If these α and β probabilistic magnitudes are just real numbers, just imagine:

- $\alpha = 1/\sqrt{2}$
- $\beta = 1/\sqrt{2}$

Then:

- $(1/\sqrt{2})^2 = 1/2$
- $(1/\sqrt{2})^2 = 1/2$
- $1/2 + 1/2 = 1$

 **Lecturer:** "That means this is an acceptable state. Now that indicates that this qubit is a combination of the $|0\rangle$ state and the $|1\rangle$ state in equal proportion. So the part of $|1\rangle$ and the part of $|0\rangle$ is the same, with no phase difference. There is no phase difference. So in this vector, $|1\rangle$ is exactly the same—whatever its primitive nature is—there is no phase difference."

EXAMPLE: COMPLEX-NUMBER PROBABILISTIC AMPLITUDES

Now, if I will introduce, instead of α , if I will give you some kind of $a_1 + b_1 i$ (this is the magnitude regarding α) and $a_2 - b_2 i$ (this is another corresponding to β), this is a real complex number—real complex numbers as a probabilistic magnitude, probabilistic amplitude.

Now this is representing:

- **Magnitude of α :** $\sqrt{a_1^2 + b_1^2}$
- **Magnitude of β :** $\sqrt{a_2^2 + b_2^2}$

These magnitudes are the **probability magnitudes**.

 **Lecturer:** "So with this probability, you will get $|1\rangle$. With this probability, you will get $|0\rangle$."

Now, $a_1 + b_1 i$ has some phase as well. What is that phase?

- **Phase:** $\tan^{-1}(b_1 / a_1)$

👤 **Lecturer:** "That is the phase. That's the phase. Now, this $|1\rangle$, which is going in the resultant vector, is not the same $|1\rangle$. It has a phase difference of this much. That's the interpretation. That's the interpretation: that this $|1\rangle$ which is making up this resultant state is not the exactly same $|1\rangle$ we are introducing."

中文全文翻译

5. 核心概念：复数作为概率幅

5.1 复数的双重角色

👤 **讲师:** "所以现在你理解了关于极坐标的陈述吗？关于你如何用极坐标表示任何复数？你是用它的长度来写的，并且那里有一个角度。所以一个复数本质上就是同时表示两个东西：一个是模长，另一个是空间。"

在自然界中，当我们需要一个单一数字来表示两个量时——例如，当我需要用单一数字来表示两个矩阵时——我就会使用复数。这是因为它能同时给我：

- 模长
- 相位差

现实世界类比：电信号

👤 **讲师:** "在电信号中，你需要一个模长和一个相位差，所以复数在那里被大量使用。同样地，在我们的量子计算案例中，我需要表示——对于那些状态——会有一个与之相关的模长，也会有一个相位差。"

5.2 为什么量子计算需要复数

👤 **讲师:** "这就是为什么我在你们面前的主张是：这就是为什么我们要理解复数，并且大量使用它们来表示我们的概率幅。因为那个概率幅不仅会向我揭示模长——它直接与概率相关——它还会给我并指示我相位差是什么。明白我的意思吗？相位差是什么？不明白？好的。"

5.3 量子比特状态表示

所以这是我的 $\alpha|1\rangle + \beta|0\rangle$ 。什么是主要状态？构成一个量子比特的主要状态是什么？

一个量子比特是 $|1\rangle$ 和 $|0\rangle$ 向量的组合。这两个向量以某种方式构成了量子比特。而这个"某种方式"是什么？

- α 和 β 是概率模长
- α 和 β 是概率幅

示例：实数概率幅

如果这些 α 和 β 概率模长只是实数，想象一下：

- $\alpha = 1/\sqrt{2}$
- $\beta = 1/\sqrt{2}$

那么：

- $(1/\sqrt{2})^2 = 1/2$
- $(1/\sqrt{2})^2 = 1/2$
- $1/2 + 1/2 = 1$

👤 讲师："这意味着这是一个可接受的状态。现在这表明这个量子比特是 $|0\rangle$ 状态和 $|1\rangle$ 状态的等比例组合。所以 $|1\rangle$ 的部分和 $|0\rangle$ 的部分是相同的，没有相位差。没有相位差。所以在这个向量中， $|1\rangle$ 是完全相同的——无论其原始性质是什么——没有相位差。"

示例：复数概率幅

现在，如果我引入，代替 α ，如果我给你某种 $a_1 + b_1 i$ （这是关于 α 的模长）和 $a_2 - b_2 i$ （这是对应于 β 的另一个），这是一个真正的复数——真正的复数作为概率模长、概率幅。

现在这表示：

- α 的模长： $\sqrt{(a_1^2 + b_1^2)}$
- β 的模长： $\sqrt{(a_2^2 + b_2^2)}$

这些模长就是概率模长。

👤 讲师："所以以这个概率，你会得到 $|1\rangle$ 。以这个概率，你会得到 $|0\rangle$ 。"

现在， $a_1 + b_1 i$ 也有一个相位。那个相位是什么？

- 相位: $\tan^{-1}(b_1 / a_1)$

👤 讲师: "那就是相位。那就是相位。现在, 这个进入结果向量的 $|1\rangle$, 并不是原来的那个 $|1\rangle$ 。它有一个这么大的相位差。这就是解释。这就是解释: 构成这个结果状态的 $|1\rangle$ 并不是我们引入的那个完全相同的 $|1\rangle$ 。"

Structured Document: Quantum Computing Lecture (Level 9 Master of Software Engineering)

Part 1: Phase Interpretation and Complex Numbers

PHASE OF A COMPLEX COEFFICIENT

The coefficient $a_1 + b_1 i$ has a phase associated with it.

- **Phase:** $\tan^{-1}(b_1 / a_1)$

👤 **Lecturer:** "That is the phase. That's the phase. Now, this one, which is going in the resultant vector, is not the same one. It has a phase difference of this much. That's the interpretation. That's the interpretation: that this one which is making up this resultant state is not the exactly same one we are introducing."

PHASE DIFFERENCE IN QUANTUM STATES

- We are introducing a **phase difference** into the $|1\rangle$ component.
- Magnitude-wise, it is still 1 (unit magnitude).
- However, it now carries a **phase difference**.

👤 **Lecturer:** "It is still one, magnitude-wise it is one, but having a phase difference in it. Is it making sense?"

WHY COMPLEX NUMBERS MATTER IN QUANTUM COMPUTING

👤 **Lecturer:** "So that's why these magnitude mostly in nature, when we will use it for an algorithmic purpose, these all will be complex numbers. And complex numbers can

represent two things: one is its magnitude, and another is the phase difference."

Key Insight:

- Complex numbers encode **two pieces of information**: magnitude and phase.
 - When we extend to any other three-dimensional number system, the concept will remain the same.
 - The base or fundamental aspect will remain consistent across different number systems.
-

Part 2: Course Pacing and Clarity Assurance

INSTRUCTOR'S REASSURANCE

 **Lecturer:** "So for today, this part is enough, otherwise it will make you confused. So whatever you have understood, that's fine. Whatever not, that is also fine. We'll make sure that in next one or two lectures, whatever is the complicated stuff, in an easy way, you will be able to understand."

Upcoming Plan:

- In the next two lectures, the instructor will continue framing this material for much more clarity.
 - Examples will be provided to reinforce understanding.
-

Part 3: Bloch Sphere Visualization (Practical Demonstration)

OPENING THE RESOURCE FILE

 **Lecturer:** "Now we will go to a word. This is 46. Yes, we will now, can you all open up this file? It will be under week 2, as well as in week 3 you will find it."

CODE IMPORTS AND SETUP

 **Lecturer:** "So there are certain imports, same sort of an import which we have done earlier as well."

PLOTTING THE BLOCH SPHERE

 **Lecturer:** "So here I am floating, if you can notice. So we are floating here. Which vector? So here, what is your expectation? What's going to come up?"

Code Execution:

```
# Plot Bloch vector with Cartesian coordinates
# We are giving three coordinates: x, y, and z
# x = 0, y = 0, z = 1
```

BLOCH SPHERE RECAP

 **Lecturer:** "Bloch sphere, I am a part of the Bloch sphere and this is the center."

Key Coordinates on the Bloch Sphere:

- **Positive z-axis:** corresponds to the $|0\rangle$ vector (north pole)
- **Negative z-axis:** corresponds to the $|1\rangle$ vector (south pole)
- These are the **primitive vectors**

 **Lecturer:** "And anything on this Bloch sphere would be a combination of these two vectors."

PLOTTING THE $|0\rangle$ VECTOR

 **Lecturer:** "So going ahead here, if you will see 0, 0, 1. So this vector is corresponding to? Up north. So it's a north pole vector, which is nothing but a 0 vector. So let me plot this."

Visualization Result:

- Coordinates: $(0, 0, 1)$
- This is the $|0\rangle$ vector pointing upward
- Axes: x, y, and z
- This is the **primitive vector** $|0\rangle$

PLOTTING THE $|1\rangle$ VECTOR

 **Lecturer:** "Now the next primitive vector is... z is what? Minus one. So zero, zero, and minus one. Down."

Visualization Result:

- Coordinates: $(0, 0, -1)$
- This is the $|1\rangle$ vector pointing downward

STUDENT TASK

 **Lecturer:** "There are certain reflections that you need to cover up and complete later. Which vector it will plot?"

中文全文翻译

第一部分：相位解释与复数

复系数的相位

系数 $a_1 + b_1 i$ 也有一个相位。

- 相位: $\tan^{-1}(b_1 / a_1)$

 **讲师:** "那就是相位。那就是相位。现在，这个进入结果向量的 $|1\rangle$ ，并不是原来的那个 $|1\rangle$ 。它有一个这么大的相位差。这就是解释。这就是解释：构成这个结果状态的 $|1\rangle$ 并不是我们引入的那个完全相同的 $|1\rangle$ 。"

量子态中的相位差

- 我们在 $|1\rangle$ 分量中引入了一个相位差。
- 从幅度上看，它仍然是 1（单位幅度）。
- 然而，它现在带有一个相位差。

 **讲师:** "从幅度上看它仍然是1，但它带有相位差。这能理解吗？"

为什么复数在量子计算中很重要

 讲师："所以这就是为什么这些幅度在自然界中，当我们将其用于算法目的时，所有这些都将是复数。复数可以表示两样东西：一个是它的幅度，另一个是相位差。"

关键见解：

- 复数编码了**两条信息**：幅度和相位。
 - 当我们扩展到任何其他三维数系时，概念将保持不变。
 - 基础或根本的方面在不同数系中保持一致。
-

第二部分：课程进度与清晰度保证

讲师的安抚

 讲师："所以今天这部分就足够了，否则会让你们感到困惑。所以无论你理解了什么，那都没问题。没理解的也没关系。我们会在接下来的一两节课中，确保所有复杂的内容都能以简单的方式让你们理解。"

后续计划：

- 在接下来的两节课中，讲师将继续构建这些内容，使其更加清晰。
 - 将提供示例来加强理解。
-

第三部分：布洛赫球可视化（实践演示）

打开资源文件

 讲师："现在我们来查看一个词。这是46。是的，我们现在，你们能打开这个文件吗？它会在第二周，以及第三周里找到。"

代码导入与设置

 讲师："这里有一些导入，和我们之前做过的导入类似。"

绘制布洛赫球

👤 讲师: "这里我在浮动, 如果你们能注意到的话。我们在这里浮动。哪个向量? 那么这里, 你们的预期是什么? 会出现什么?"

代码执行:

```
# 用笛卡尔坐标绘制布洛赫向量  
# 我们给出三个坐标: x、y 和 z  
#  $x = 0$ ,  $y = 0$ ,  $z = 1$ 
```

布洛赫球回顾

👤 讲师: "布洛赫球, 我是布洛赫球的一部分, 这是中心。"

布洛赫球上的关键坐标:

- **正z轴**: 对应 $|0\rangle$ 向量 (北极)
- **负z轴**: 对应 $|1\rangle$ 向量 (南极)
- 这些是**原始向量**

👤 讲师: "布洛赫球上的任何东西都将是这两个向量的组合。"

绘制 $|0\rangle$ 向量

👤 讲师: "继续往下看, 如果你们看到 0, 0, 1。那么这个向量对应的是? 北极。所以它是一个北极向量, 也就是 $|0\rangle$ 向量。让我来绘制它。"

可视化结果:

- 坐标: $(0, 0, 1)$
- 这是指向上的 $|0\rangle$ 向量
- 坐标轴: x、y 和 z
- 这是**原始向量** $|0\rangle$

绘制 $|1\rangle$ 向量

👤 讲师: "现在下一个原始向量是.....z 是多少? 负一。所以是 0, 0, 和负一。向下。"

可视化结果：

- 坐标： $(0, 0, -1)$
- 这是指向下的 $|1\rangle$ 向量

学生任务

 讲师： "有一些反射需要你们自己去完成和补充。它会绘制哪个向量？ "

Of course. Here is the processed content based on the provided subtitle text, following all specified requirements.

1. Zero Information Loss & Structural Reorganization (English Only)

VISUALIZING PRIMITIVE VECTORS ON THE BLOCH SPHERE

The lecture continues with the visualization of primitive vectors on the Bloch sphere.

- **The $|0\rangle$ Vector:** The previous segment established the $|0\rangle$ vector pointing up along the positive z-axis.
- **The $|1\rangle$ Vector:** The next primitive vector is $|1\rangle$.
 - Its z-coordinate is -1.
 - Its coordinates are $(0, 0, -1)$.
 - This vector points **down** along the negative z-axis.

STUDENT INTERACTION: IDENTIFYING A VECTOR ON THE EQUATOR

 **Lecturer:** "There are certain reflections that you need to cover up and complete later. Which vector it will plot?"

 **Lecturer:** "This one."

 **Student:** "The y."

 **Lecturer:** "Yes."

 **Lecturer:** "What will be the vector plot?"

 **Student:** "Positive y axis."

 **Lecturer:** "Positive y axis so it will be on equator and that's a combination of 0 and 1 and equally."

THE EQUATOR AND EQUAL SUPERPOSITION

The lecturer explains the significance of vectors lying on the equator of the Bloch sphere.

- Any vector on the **equator plane** has an equal probability of collapsing to either $|0\rangle$ or $|1\rangle$ upon measurement.
- The state vector for any such point is: $|\psi\rangle = (1/\sqrt{2}) |0\rangle + (1/\sqrt{2}) |1\rangle$
- The probabilistic amplitudes for both $|0\rangle$ and $|1\rangle$ are $1/\sqrt{2}$.
- This state is always in **equal superposition**.

THE HADAMARD GATE AND PHASE DIFFERENCE

The lecturer discusses the effect of the **Hadamard gate** on a vector on the equator.

- If you apply the Hadamard gate to a vector on the equator, it will fall onto the **X-axis**.
- In this specific case, there is **no introduced phase difference** because the vector is exactly $(1/\sqrt{2}) |0\rangle + (1/\sqrt{2}) |1\rangle$.
- However, for other vectors on the equator, the result will not be the same. It will have some kind of a **phase difference** somewhere in their probabilistic magnitude.

SPHERICAL COORDINATES ON THE BLOCH SPHERE

The lecturer transitions to a discussion of spherical coordinates, which is fundamental for understanding the Bloch sphere.

 **Lecturer:** "Spherical coordinate so this is important for us. Can you recall the Bloch sphere explanation which I gave you last time? What are these angles? Where the vector will end up using this? Where the vector will end up? So last time I gave you some explanation regarding spherical coordinate on Bloch sphere. Can you name those two angles? Can you name those two angles? What were the names of those two angles? Polar angle, another was... I do meet the lander. So what is polar angle?"

The lecturer then draws a diagram on the board.

 **Lecturer:** "That's me the x is this angle between the x's so I'm just executing it and I'm leaving it for your interpretation. Find for me what is polar angle in it and what is the azimuthal angle in it. So this is the final result for it. So tell me, take some time okay. This is fundamental and important. If you will skip this... yes."

STUDENT INTERACTION: IDENTIFYING THE POLAR ANGLE

 **Lecturer:** "What is polar angle in it and what is azimuthal angle in it? What is polar angle? Polar angle is in between positive z axis... x is this one. What is an angle equation? 90. So that is $\pi/2$. So the first angle when we are floating the plot..."

2. Full Chinese Translation (中文全文翻译)

在布洛赫球面上绘制原始向量

课程继续讲解如何在布洛赫球面上可视化原始向量。

- $|0\rangle$ 向量：上一段已经确定 $|0\rangle$ 向量沿正 z 轴向上。
- $|1\rangle$ 向量：下一个原始向量是 $|1\rangle$ 。
 - 它的 z 坐标是 -1。
 - 它的坐标是 $(0, 0, -1)$ 。
 - 这个向量沿负 z 轴向下。

师生互动：识别赤道上的向量

 讲师："有一些反射需要你们自己去完成和补充。它会绘制哪个向量？"

 讲师："这个。"

 学生："y 轴。"

 讲师："是的。"

 讲师："它会绘制哪个向量？"

 学生："正 y 轴。"

👤 讲师: "正 y 轴, 所以它会在赤道上, 并且是 0 和 1 的等量组合。"

赤道与等概率叠加

讲师解释了位于布洛赫球赤道上的向量的意义。

- 赤道平面上的任何向量在测量时, 坍缩到 $|0\rangle$ 或 $|1\rangle$ 的概率相等。
- 任何此类点的状态向量为: $|\psi\rangle = (1/\sqrt{2}) |0\rangle + (1/\sqrt{2}) |1\rangle$
- $|0\rangle$ 和 $|1\rangle$ 的概率幅都是 $1/\sqrt{2}$ 。
- 这种状态始终处于**等概率叠加**。

哈达玛门与相位差

讲师讨论了将哈达玛门应用于赤道上的向量的效果。

- 如果你对赤道上的一个向量应用哈达玛门, 它会落到 **X 轴**上。
- 在这个特定情况下, **没有引入相位差**, 因为这个向量恰好是 $(1/\sqrt{2}) |0\rangle + (1/\sqrt{2}) |1\rangle$ 。
- 然而, 对于赤道上的其他向量, 结果会不同。它们的概率幅中会存在某种**相位差**。

布洛赫球面上的球坐标

讲师转向讨论球坐标, 这对于理解布洛赫球面至关重要。

👤 讲师: "球坐标, 这对我们很重要。你们能回忆起我上次给出的关于布洛赫球面的解释吗? 这些角度是什么? 使用这个, 向量最终会落在哪里? 向量最终会落在哪里? 上次我给出了一些关于布洛赫球面上球坐标的解释。你们能说出那两个角的名字吗? 你们能说出那两个角的名字吗? 那两个角的名字是什么? 极角, 另一个是.....方位角。那么什么是极角?"

讲师随后在黑板上画了一个图。

👤 讲师: "那是我的 x 轴, 这是 x 轴之间的角度, 所以我只是执行它, 然后留给你们自己去解读。帮我找出其中的极角和方位角是什么。这就是它的最终结果。告诉我, 花点时间, 好的。这是基础且重要的。如果你们跳过这个.....是的。"

师生互动: 识别极角

👤 讲师: "其中的极角是什么, 方位角是什么? 什么是极角? 极角是在正 z 轴和.....x 轴是这个。角度方程是什么? 90。所以是 $\pi/2$ 。所以当我们绘制图形时, 第一个角度是....."

好的, 作为您的学术助理, 我将严格遵循您的5项要求, 对这份Level 9 Quantum Computing 课程字幕文件进行处理。

1. Zero Information Loss & Structural Reorganization (English Only)

IDENTIFYING POLAR AND AZIMUTHAL ANGLES

The lecturer continues the discussion on spherical coordinates, focusing on identifying the **polar angle** and **azimuthal angle** for a given vector.

- The **polar angle** is the angle between the vector and the positive z-axis.
 - It can vary from 0 to 180 degrees.
 - A polar angle of 0 degrees means the vector points directly along the positive z-axis.
 - A polar angle of 180 degrees means the vector points directly along the negative z-axis.
 - A polar angle of 90 degrees means the vector lies in the xy-plane.
- The **azimuthal angle** is the angle of the vector's projection onto the xy-plane, measured from the positive x-axis.
 - It can vary from 0 to 360 degrees.
 - Imagine a light source shining from above, parallel to the z-axis. The shadow of the vector on the xy-plane makes an angle with the positive x-axis; that is the azimuthal angle.

EXAMPLE: A VECTOR IN THE XY-PLANE

The lecturer uses a specific vector as an example.

- The vector lies in the xy-plane.
- **Polar angle:** Since the vector is in the xy-plane, the polar angle is 90 degrees, or $\pi/2$.
- **Azimuthal angle:** The vector makes an angle of 60 degrees with the positive x-axis. Therefore, the azimuthal angle is 60 degrees, or $\pi/3$.

STUDENT INTERACTION: CONFIRMING THE AZIMUTHAL ANGLE

🧑‍🎓 **Student:** "Yeah, pi by 3." 🧑‍🎓 **Lecturer:** "Pi by 3, that is. So pi is 180 degree. 180 degree by 3 is 60 degree. So that is a 60 degree angle. So this angle is 60 degree."

EXERCISE: PLOTTING A VECTOR ON THE NEGATIVE Y-AXIS

The lecturer then sets a new exercise: modify the previous cell to draw a vector pointing along the negative y-axis.

- In Cartesian coordinates, this is simple: $x=0, y=-1, z=0$.
- In spherical coordinates, you must find the correct **polar angle (θ)** and **azimuthal angle (φ)**.

STUDENT INTERACTION: SOLVING THE EXERCISE

🧑‍🎓 **Lecturer:** "Find it out. Yes, what will be your polar coordinate, means polar angle for that. Still it will be?" 🧑‍🎓 **Lecturer:** "90." 🧑‍🎓 **Lecturer:** "But what will be your?" 🧑‍🎓 **Student:** "3 by 3." 🧑‍🎓 **Student:** "3 pi by 2." 🧑‍🎓 **Lecturer:** "Because if you will see from here 90 degree, 90 degree and 90 degree, 3 times 90 degree. So 3 pi by 2 that means 270 degree. So make an entry there in arguments. first angle is same, pi by 2, but the third angle will be 3 pi by 2. The third argument will be 3 pi by 2."

CONCLUSION ON VISUALIZATION

The lecturer concludes that this type of visualization is critical for understanding quantum states.

2. Full Chinese Translation (中文全文翻译)

识别极角和方位角

讲师继续讨论球坐标，重点是如何为给定的向量识别**极角**和**方位角**。

- **极角**是向量与正z轴之间的夹角。
 - 它的取值范围是0到180度。
 - 极角为0度意味着向量直接指向正z轴方向。

- 极角为180度意味着向量直接指向负z轴方向。
- 极角为90度意味着向量位于xy平面内。
- **方位角**是向量在xy平面上的投影与正x轴之间的夹角。
 - 它的取值范围是0到360度。
 - 想象一束从上方平行于z轴照射的光源。向量在xy平面上的阴影与正x轴形成一个夹角，这个角就是方位角。

示例：xy平面内的一个向量

讲师使用一个具体的向量作为例子。

- 该向量位于xy平面内。
- **极角**：由于向量在xy平面内，极角为90度，即 $\pi/2$ 。
- **方位角**：该向量与正x轴成60度角。因此，方位角为60度，即 $\pi/3$ 。

师生互动：确认方位角

🧑 学生：“是的， $\pi/3$ 。” 🧑 讲师：“ $\pi/3$ ，没错。 π 是180度。180度除以3是60度。所以这是一个60度的角。这个角就是60度。”

练习：绘制指向负y轴的向量

讲师随后布置了一个新的练习：修改之前的代码单元，绘制一个指向负y轴的向量。

- 在笛卡尔坐标系中，这很简单： $x=0, y=-1, z=0$ 。
- 在球坐标系中，你必须找到正确的**极角 (θ)** 和**方位角 (ϕ)**。

师生互动：解答练习

🧑 讲师：“把它找出来。是的，它的极坐标是什么，也就是极角是多少？它仍然是……？”

🧑 讲师：“90。” 🧑 讲师：“但是你的……是多少？” 🧑 学生：“ $3/3$ 。” 🧑 学生：“ $3\pi/2$ 。”

🧑 讲师：“因为如果你从这里看，90度，90度，90度，3倍的90度。所以是 $3\pi/2$ ，也就是270度。所以在参数那里输入。第一个角度相同，是 $\pi/2$ ，但第三个角度将是 $3\pi/2$ 。第三个参数将是 $3\pi/2$ 。”

关于可视化的结论

讲师总结道，这种类型的可视化对于理解量子态至关重要。

Quantum Computing Lecture - Continued: Bloch Sphere Visualization

H2: Visualizing the $|1\rangle$ State on the Bloch Sphere

H3: DETERMINING THETA AND PHI FOR THE $|1\rangle$ VECTOR

The lecturer continues the discussion on spherical coordinates for quantum state visualization.

 **Lecturer:** "So one vector, if you have to draw one vector using spherical coordinate system, what will be your theta and what will be your phi?"

 **Student:** "Theta has to be zero, right?"

 **Lecturer:** "And then phi will be?"

 **Student:** "Theta is zero."

The lecturer corrects this understanding:

 **Lecturer:** "Theta is 180 degrees. Theta will be 180 degrees. What will be phi?"

The lecturer explains the reasoning:

- **Definition of phi:** You have to find its projection on the xy-plane.
- For the $|1\rangle$ vector, the projection on the xy-plane will be just a point.
- Therefore, the angle (phi) will be **zero**.

H3: THE CRITICAL IMPORTANCE OF BLOCH SPHERE VISUALIZATION

The lecturer emphasizes that this visualization is fundamental:

 **Lecturer:** "So this visualization is something critical. You have to be habitual of this visualization if you have to continue with me. This diagram is important and visualizing it for any vector, any random vector—if I will give you, you should have visualization. Okay, this will be its theta and that will be its phi."

Key points about single-qubit visualization:

- For a **single qubit**, you can visualize that qubit through the **Bloch sphere**.
- Whatever manipulations are happening on that qubit can be visualized using the Bloch sphere.
- This will be quite helpful for understanding each and every operator.

H3: UPCOMING TOPICS - GATE OPERATIONS ON THE BLOCH SPHERE

The lecturer outlines the pedagogical approach:

1. **First:** We will have a Bloch sphere *without* any operator applied.
2. **Then:** We will have a Bloch sphere *with* an operator applied on the qubit.
 - This includes gates like: **Hadamard gate**, **X gate**, **Y gate**, **Z gate**, and all sorts of gates.
3. **Result:** You will see how vector manipulation (qubit manipulation) happens.
4. **Once this visualization is understood:** It will be much easier to bring two or three qubits together and see how they can fit into an algorithm to solve our problems.

 **Lecturer:** "Visualization is important here. So single qubit, if you have to visualize, this is the tool to visualize it."

H3: FUTURE MULTI-QUBIT VISUALIZATION

The lecturer acknowledges that visualization tools will evolve:

- When **multiple qubits** come into play, there are multiple other tools available.
- "Maybe in next or whenever it will occur to us, if we need to learn those kind of tools, we will learn those as well."

H3: TERMINOLOGY CHECK

 **Lecturer:** "But first please, azimuthal angle and?"

 **Student:** "Polar angle."

 **Lecturer:** "Polar angle. So polar angle will give us an indication that how close our vector is with respect to..."

H2：在布洛赫球面上可视化 $|1\rangle$ 态

H3：确定 $|1\rangle$ 向量的 θ 和 ϕ

讲师继续讨论用于量子态可视化的球坐标。

👤 讲师："那么，如果你要用球坐标系画一个向量，对于 $|1\rangle$ 向量，你的 θ 会是多少，你的 ϕ 会是多少？"

👤 学生：" θ 应该是零，对吗？"

👤 讲师："那 ϕ 呢？"

👤 学生：" θ 是零。"

讲师纠正了这个理解：

👤 讲师：" θ 是 180 度。 θ 会是 180 度。 ϕ 会是多少？"

讲师解释了推理过程：

- ϕ 的定义：你需要找到它在 xy 平面上的投影。
- 对于 $|1\rangle$ 向量，它在 xy 平面上的投影将只是一个点。
- 因此，那里的角度 (ϕ) 将是 零。

H3：布洛赫球面可视化的关键重要性

讲师强调这种可视化是基础性的：

👤 讲师："所以这种可视化是至关重要的。如果你要继续跟我学下去，你必须习惯这种可视化。这个图很重要，对于任何向量，任何随机向量——如果我给你一个，你应该能可视化它。好的，这是它的 θ ，那是它的 ϕ 。"

关于单量子比特可视化的要点：

- 对于单个量子比特，你可以通过布洛赫球面来可视化该量子比特。
- 对该量子比特发生的任何操作都可以使用布洛赫球面进行可视化。
- 这将对理解每一个算符非常有帮助。

H3：即将讨论的主题——布洛赫球面上的门操作

讲师概述了教学方法：

1. **首先**：我们将有一个没有应用任何算符的布洛赫球面。
2. **然后**：我们将有一个在量子比特上应用了算符的布洛赫球面。
 - 这包括诸如：**哈达玛门**、**X 门**、**Y 门**、**Z 门**以及所有类型的门。
3. **结果**：你将看到向量操作（量子比特操作）是如何发生的。
4. **一旦理解了这种可视化**：将更容易把两个或三个量子比特放在一起，看看它们如何能融入算法并解决我们的问题。

 **讲师**："可视化在这里很重要。所以对于单个量子比特，如果你需要可视化，这就是可视化的工具。"

H3：未来的多量子比特可视化

讲师承认可视化工具将会发展：

- 当涉及**多个量子比特**时，还有其他多种工具可用。
- "也许在下一节课，或者当我们遇到需要的时候，如果需要学习那些工具，我们也会学习它们。"

H3：术语确认

 **讲师**："但首先请问，方位角是？"

 **学生**："极角。"

 **讲师**："极角。所以极角会告诉我们，我们的向量相对于...有多接近。"

好的，作为您的严谨学术助理，我已根据您的要求，对提供的字幕文件进行了处理。

Level 9 Master of Software Engineering - Quantum Computing Course

H1: LECTURE 1 - VISUALIZING QUBITS ON THE BLOCH SPHERE (CONTINUED)

H2: Understanding the Bloch Sphere Angles

- The **polar angle (θ)** indicates how close the vector is to the **North Pole** or the **South Pole**.
 - If the vector is near the **South Pole**, there is a higher probability that when it collapses, it will collapse to $|1\rangle$.
 - If the vector is near the **North Pole**, it is more likely to collapse to $|0\rangle$.
- The **azimuthal angle (ϕ)** indicates the rotation around the Z-axis.
 - It measures how much the vector has deviated from the X-axis on the XY-plane.
 - This rotation ranges from 0 to 360 degrees.

 **Lecturer:** "But first please, azimuthal angle and?"

 **Student:** "Polar angle."

 **Lecturer:** "Polar angle. So polar angle will give us an indication that how close our vector is with respect to North Pole or how close the vector is with respect to South Pole."

- The **polar angle** ranges from 0 to 180 degrees.
- The **azimuthal angle** ranges from 0 to 360 degrees.

H2: Coordinate Conversions

- The lecturer explains the conversion from **Cartesian coordinates** to **polar coordinates** and vice-versa.
- An example is given for a vector with $\theta = \pi/2$ and $\phi = \pi/3$ (60 degrees).
- The conversion is described as *simple trigonometry* and is not strictly required for the course, but useful for understanding.
- The 3D nature of the Bloch sphere makes the conversion slightly more complex.

 **Lecturer:** "If theta is 0, where the vector will be?"

 **Student:** "On z axis."

 **Lecturer:** "On z axis. If the theta is pi, the vector is on?"

 **Student:** "Negative z."

H2: First Quantum Gate - The Hadamard Gate

- The lecturer introduces the first quantum gate to be applied to a qubit.

- **Initial State:** When initializing a quantum circuit, the initial qubit state is $|0\rangle$, which is located on the **positive Z-axis**.
- **Action:** The **Hadamard gate** is applied to this initial state.
- **Implementation:** The process involves:
 1. Creating an instance of a quantum circuit.
 2. Applying the Hadamard gate.
 3. Printing the circuit.
 4. Taking the state vector and plotting it.

This is a simple procedure to analyze the effect of the Hadamard gate on the initial $|0\rangle$ state.

中文全文翻译

软件工程硕士 9 级 - 量子计算课程

第一讲：在布洛赫球面上可视化量子比特（续）

理解布洛赫球面的角度

- **极角 (θ)** 表示向量距离**北极**或**南极**的接近程度。
 - 如果向量靠近**南极**，那么当它坍缩时，坍缩到 $|1\rangle$ 的概率更高。
 - 如果向量靠近**北极**，它更有可能坍缩到 $|0\rangle$ 。
- **方位角 (ϕ)** 表示绕 Z 轴的旋转。
 - 它衡量向量在 XY 平面上偏离 X 轴的程度。
 - 这个旋转范围是 0 到 360 度。

👤 讲师：“但首先请问，方位角是？”

👤 学生：“极角。”

👤 讲师：“极角。所以极角会告诉我们，我们的向量相对于北极有多近，或者相对于南极有多近。”

- **极角**的范围是 0 到 180 度。
- **方位角**的范围是 0 到 360 度。

坐标转换

- 讲师解释了从笛卡尔坐标到极坐标以及反向的转换。
- 给出了一个向量示例，其 $\theta = \pi/2$ ， $\phi = \pi/3$ (60 度)。
- 这种转换被描述为简单的三角学，并非课程严格必需，但有助于理解。
- 布洛赫球的 3D 性质使得转换稍微复杂一些。

👤 讲师：“如果 theta 是 0，向量会在哪里？”

👤 学生：“在 z 轴上。”

👤 讲师：“在 z 轴上。如果 theta 是 pi，向量会在哪里？”

👤 学生：“负 z 轴。”

第一个量子门 - 哈达玛门

- 讲师介绍了将应用于量子比特的第一个量子门。
- **初始状态：** 初始化量子电路时，初始量子比特状态是 $|0\rangle$ ，它位于正 Z 轴上。
- **操作：** 哈达玛门被应用于这个初始状态。
- **实现：** 该过程包括：
 1. 创建一个量子电路的实例。
 2. 应用哈达玛门。
 3. 打印电路。
 4. 获取状态向量并绘制它。

这是一个简单的过程，用于分析哈达玛门对初始 $|0\rangle$ 状态的影响。

Quantum Computing Lecture - Continued Analysis of Quantum Gates

H2: Analyzing the Hadamard Gate on the Bloch Sphere

H3: VISUALIZING THE STATE VECTOR TRANSFORMATION

The lecturer continues the demonstration of applying the **Hadamard gate** to the initial $|0\rangle$ state:

- **Initial State:** The qubit vector was located on the **positive Z-axis** (the $|0\rangle$ state).
- **Operation:** The **Hadamard gate** is applied to this initial state.
- **Implementation Process:**
 1. Create an instance of a quantum circuit.
 2. Apply the Hadamard gate.
 3. Print the circuit.
 4. Obtain the state vector and plot it.

This is a straightforward procedure designed to analyze what manipulation occurs when the Hadamard gate is applied.

H3: RESULTS OF THE HADAMARD GATE APPLICATION

When the lecturer executes this code, the following results are observed:

- **Vector Transformation:** The qubit vector, initially on the Z-axis, undergoes a **90-degree rotation** onto the **positive X-axis** after applying the Hadamard gate.
- **State Vector Components:** The resulting state vector has the following **probabilistic amplitudes**:
 - Both amplitudes are $1/\sqrt{2}$ (approximately 0.707).
 - This is the **equal superposition state**.

Key Observation: The qubit was initially in the $|0\rangle$ state, and after applying the Hadamard gate, it rotates 90 degrees to the positive X-axis, creating a superposition state.

H3: UNDERSTANDING SUPERPOSITION STATES

The lecturer explains the nature of superposition states on the **Bloch sphere**:

- **Definition of Superposition:** Any vector that falls on the **XY-plane** of the Bloch sphere is in a **superposition state**.
- **Infinite Possibilities:** There are infinite possible vectors on the XY-plane, with a **0 to 360 degree variation**.
- **Equal vs. Unequal Superposition:**
 - **Equal Superposition:** When both amplitudes are $1/\sqrt{2}$ and are **real numbers** with **no introduced phase difference**.

- **Special Case:** The current state ($1/\sqrt{2}$ and $1/\sqrt{2}$) is a special superposition because:
 - Both amplitudes are real numbers.
 - There is **no phase difference** introduced between the $|0\rangle$ and $|1\rangle$ components.
 - The combination of 0 and 1 is **intact** (in-phase).
- **Other XY-plane Vectors:** Other vectors on the XY-plane will also be in superposition, but:
 - The amplitudes for $|0\rangle$ and $|1\rangle$ will **not be equal**.
 - They will have an **introduced phase difference**.
 - Magnitude-wise, the probabilities may be the same, but the **phase difference changes**.

H2: Introduction to the X Gate (NOT Gate)

H3: WHAT IS THE X GATE?

The lecturer transitions to discussing the **X gate**:

- **Definition:** The **X gate** is essentially a **NOT gate** in quantum computing.
- **Why the Name "X"?** The name comes from the axis of rotation:
 - A **NOT gate** introduces a **180-degree phase difference**.
 - The **X gate** introduces a **180-degree phase difference**.
 - This rotation happens with respect to the **X-axis**.

Key Point: The X gate performs a 180-degree rotation around the X-axis on the Bloch sphere.

 **Student:** (No specific student question recorded in this segment)

 **Lecturer:** "So earlier the qubit was like this, and then the qubit will turn 90 degrees to the bigger... and this state is last time we understood the superposition state. So this is the superposition state. So any vector which will fall on XY plane, whenever they will land on XY plane, they are in equal superposition. Wherever, so there are infinite possibilities. So you have 0 to 360 degree variation, so there will be numerous vectors possible which may be in superposition. Right now this is a special superposition because the $1/\sqrt{2}$ and $1/\sqrt{2}$ both are real numbers, and there is no introduced phase difference. So both combination of 0 and 1 are the intact combinations. So there is no introduced phase difference into them."

中文全文翻译

H2: 在布洛赫球上分析哈达玛门

H3: 可视化状态向量变换

讲师继续演示将哈达玛门应用于初始 $|0\rangle$ 状态的过程：

- **初始状态：** 量子比特向量位于正 **Z** 轴上 ($|0\rangle$ 状态)。
- **操作：** 哈达玛门被应用于这个初始状态。
- **实现过程：**
 1. 创建一个量子电路的实例。
 2. 应用哈达玛门。
 3. 打印电路。
 4. 获取状态向量并绘制它。

这是一个简单的过程，旨在分析应用哈达玛门时会发生什么样的操作。

H3: 哈达玛门应用的结果

当讲师执行这段代码时，观察到以下结果：

- **向量变换：** 最初位于 Z 轴上的量子比特向量，在应用哈达玛门后，发生了**90 度旋转**，到达了正 **X** 轴上。
- **状态向量分量：** 最终的状态向量具有以下**概率幅**：
 - 两个概率幅都是 $1/\sqrt{2}$ (约 0.707)。
 - 这就是**等幅叠加态**。

关键观察： 量子比特最初处于 $|0\rangle$ 状态，应用哈达玛门后，它旋转 90 度到正 X 轴，从而创建了一个叠加态。

H3: 理解叠加态

讲师解释了布洛赫球上叠加态的性质：

- **叠加态的定义：** 任何落在布洛赫球**XY 平面**上的向量都处于**叠加态**。
- **无限可能性：** XY 平面上存在无限多个可能的向量，具有 **0 到 360 度**的变化。

- 等幅叠加与非等幅叠加：
 - 等幅叠加：当两个概率幅都是 $1/\sqrt{2}$ ，并且都是实数，且没有引入相位差时。
 - 特殊情况：当前状态（ $1/\sqrt{2}$ 和 $1/\sqrt{2}$ ）是一种特殊的叠加态，因为：
 - 两个概率幅都是实数。
 - $|0\rangle$ 和 $|1\rangle$ 分量之间没有引入相位差。
 - 0 和 1 的组合是完整的（同相）。
- 其他 XY 平面向量：XY 平面上的其他向量也处于叠加态，但是：
 - $|0\rangle$ 和 $|1\rangle$ 的概率幅不相等。
 - 它们会有一个引入的相位差。
 - 从幅度（概率）上看可能相同，但相位差发生了变化。

H2: X 门（非门）介绍

H3: 什么是 X 门？

讲师过渡到讨论 X 门：

- 定义：在量子计算中，X 门本质上就是一个非门。
- 为什么叫 "X"？这个名字来源于旋转轴：
 - 非门引入一个 **180 度** 的相位差。
 - X 门引入一个 **180 度** 的相位差。
 - 这个旋转是围绕 **X 轴** 进行的。

关键点：X 门在布洛赫球上执行围绕 X 轴的 180 度旋转。

 学生：（此段未记录具体学生问题）

 讲师："所以之前的量子比特是这样的，然后量子比特会旋转 90 度到更大的位置.....这个状态就是我们上次理解的叠加态。所以这就是叠加态。所以任何落在 XY 平面上的向量，只要它们落在 XY 平面上，就处于等幅叠加态。无论在哪里，都有无限的可能性。所以你有 0 到 360 度的变化，所以会有无数可能的向量处于叠加态。现在这是一个特殊的叠加态，因为 $1/\sqrt{2}$ 和 $1/\sqrt{2}$ 都是实数，并且没有引入相位差。所以 0 和 1 的组合都是完整的组合。它们之间没有引入相位差。"

好的，作为您的学术助理，我已经严格按照您的要求，对这份 Level 9 Master of Software Engineering 级别的 Quantum Computing 课程视频字幕文件进行了处理。

1. ZERO INFORMATION LOSS & STRUCTURAL REORGANIZATION (ENGLISH ONLY)

H2: The X Gate and Phase Difference

The lecturer explains the naming convention of the X gate. The name "X" is derived from the axis of rotation. The gate introduces a 180-degree phase difference, and this rotation occurs with respect to the X-axis.

- The X gate is essentially a NOT gate.
- "NOT" refers to the 180-degree phase difference it introduces.
- The rotation for this phase difference happens with respect to the X-axis.
- Applying the X gate converts the $|0\rangle$ vector into the $|1\rangle$ vector.

H2: Visualizing the X Gate on a Circuit

The lecturer demonstrates a simple circuit with a single qubit, applying the X gate. The state vector is then plotted.

 **Lecturer:** "So this is the circuit, simply single qubit X gate we are applying. Then whatever the state vector is, we are just plotting the state vector. And this is a state vector for this state vector. Can you write like the I have written one by root 2. Can you write the state vector for this? Can you write the state vector? For state vector, what do you need to write?"

 **Lecturer:** "So state vector will always be in this way. It's equal to alpha. You need to know the alpha. What is the zero component? The alpha component plus beta. One."

 **Lecturer:** "So what do you think? What will be the alpha? Zero. What will be beta here? One. So the vector will be? Zero, comma one. That's the same vector. That's the state vector."

H2: State Vector Notation: Bra-Ket

The lecturer explains the two primary notations for representing a state vector: the column vector and the Bra-Ket notation.

- **Column Vector:** State vectors are generally written as a column vector. For the state $|1\rangle$, this is written as:

[0]

[1]

- **Bra-Ket Notation:** This is a more compact notation used for manual calculations.
 - $|\psi\rangle$ is called the **Ket** notation.
 - $\langle\psi|$ is called the **Bra** notation.
 - The lecturer asks the class to identify the difference between the two notations.

 **Lecturer:** "So when I'm writing something like this, it's so expended form is this. This is expended form. This is just for manual thing, like when I have to do calculation, some normal humans have to do calculation. Then I will use this cat notation. This is called the cat notation. This is called the cat notation. So column vector."

 **Lecturer:** "Now same stuff, if I have to write, if I will write it like this. Sorry. Is there any difference in this and this?"

 **Student:** "Yeah."  **Student:** "One."  **Lecturer:** "No, this is one. This is also one. So can you notice a difference?"

 **Student:** "Negative."  **Lecturer:** "In notation, is there any difference? It is starting with a bar and then ending with that symbol. It is starting with less than symbol and ending with a bar. So this is called a bra notation. This is called cat. This is called bra notation."

2. FULL CHINESE TRANSLATION (中文全文翻译)

H2: X 门与相位差

讲师解释了 X 门命名的由来。名称“X”源自旋转轴。该门引入 180 度的相位差，并且这种旋转是相对于 X 轴发生的。

- X 门本质上就是一个非门。
- “非”指的是它引入的 180 度相位差。
- 这个相位差的旋转是相对于 X 轴发生的。
- 应用 X 门会将 $|0\rangle$ 向量转换为 $|1\rangle$ 向量。

H2: 在电路上可视化 X 门

讲师演示了一个简单的单量子比特电路，并应用了 X 门。然后绘制了状态向量。

👤 讲师：“所以这就是电路，我们只是应用了简单的单量子比特 X 门。然后，无论状态向量是什么，我们只是绘制出状态向量。这就是这个状态向量的状态向量。你能像我写根号二分之一那样写出来吗？你能为这个写出状态向量吗？你能写出状态向量吗？对于状态向量，你需要写什么？”

👤 讲师：“所以状态向量总是这样。它等于 α 。你需要知道 α 。零分量是什么？ α 分量加上 β 。一。”

👤 讲师：“那么你认为呢？ α 会是什么？零。这里的 β 会是什么？一。那么向量会是？零，逗号一。这就是同一个向量。这就是状态向量。”

H2: 状态向量表示法：Bra-Ket

讲师解释了表示状态向量的两种主要符号：列向量和 Bra-Ket 符号。

- **列向量**：状态向量通常写成列向量。对于 $|1\rangle$ 状态，它写成：

```
[0]
[1]
```

- **Bra-Ket 符号**：这是一种用于手动计算的更紧凑的符号。
 - $|\psi\rangle$ 被称为 **Ket** 符号。
 - $\langle\psi|$ 被称为 **Bra** 符号。
 - 讲师让班级同学识别这两种符号之间的区别。

👤 讲师：“所以当我像这样写的时候，它的展开形式是这样的。这是展开形式。这只是为了手动计算，比如当我需要进行计算时，一些普通人需要进行计算。然后我会使用这个 cat 符号。这被称为 cat 符号。这被称为 cat 符号。所以是列向量。”

👤 讲师：“现在同样的东西，如果我必须写，如果我像这样写。抱歉。这个和这个有什么区别吗？”

👤 学生：“有。” 👤 学生：“一。” 👤 讲师：“不，这是一。这也是一。你能注意到区别吗？”

👤 学生：“负的。” 👤 讲师：“在符号上，有区别吗？它以一个竖线开始，以那个符号结束。它以一个小于号开始，以一个竖线结束。所以这被称为 bra 符号。这被称为 cat。这被称为 bra 符号。”

Quantum Computing Lecture - Bra-Ket Notation & Gate Operations

H1: Bra-Ket Notation Deep Dive

H2: DISTINGUISHING BRA FROM KET

The lecturer continues the discussion on notation differences between bra and ket vectors.

 **Student:** "Negative."  **Lecturer:** "In notation, is there any difference? It is starting with a bar and then ending with that symbol. It is starting with a less-than symbol and ending with a bar. So this is called a bra notation. This is called ket. This is called bra notation."

H2: DEFINITION OF BRA NOTATION

Bra notation converts a ket vector into a **row vector**, and for any complex numbers present, it takes their **complex conjugate**.

- A ket vector is represented as a column vector
- A bra vector is represented as a row vector
- The bra is the **conjugate transpose** of the ket

H2: COMPLEX CONJUGATE EXPLANATION

 **Lecturer:** "What is complex conjugate? Any complex number, its complex conjugate is... so if the complex number is $a + bi$..."

For real numbers (where the imaginary part is zero), the complex conjugate is the same number. In this example, both numbers are real, so there is no complex part.

 **Lecturer:** "So this is?"  **Student:** "0 and 1."  **Student:** "Yeah."

H2: MATRIX MULTIPLICATION WITH BRA AND KET

H3: Dimensional Analysis

 **Lecturer:** "Now can you multiply this vector with this vector?"  **Student:** "Can you multiply? This is rho into 1, 1."  **Lecturer:** "Can you multiply this vector means this is our matrix? A simple 2×1 matrix and it is 1×2 matrix."  **Student:** "Yeah, it can be."  **Lecturer:** "So this is 1×2 matrix and it is 2×1 matrix. Can you multiply 2×1 matrix with 1×2 matrix? Yes. You can multiply because 1 and 1 is common."

H3: Inner Product (Bra-Ket)

When you multiply a **ket** (column vector) with a **bra** (row vector), you get a **scalar number**.

Example:

$$|\psi\rangle = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \langle\phi| = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$\langle\phi|\psi\rangle = \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \times 1 + 2 \times 2 = 5 \text{ (scalar)}$$

 **Lecturer:** "So it is ket dot bra. So when you will multiply this, you will get a scalar number. You will get a scalar number. You will get to [the result]."

H2: SIGNIFICANCE IN QUANTUM COMPUTING

 **Lecturer:** "So in quantum computing, it has also a significance. So what this number is, it will also reveal something. Anyways, so my intention for today is that you should understand what is ket notation and what is bra notation. So how they get used, we will understand all that later."

H1: Visualizing Gate Operations

H2: HADAMARD GATE APPLICATION

 **Lecturer:** "So now you have to just visualize then we will run the search. You have to visualize Hadamard application. So there is a zeroth vector, you are applying a Hadamard gate on it, so where the vector will end up?"

H3: Initial State Analysis

- The vector is initially on the **Z-axis**, specifically the **positive Z-axis**

 **Lecturer:** "So earlier the vector is on which axis?"  **Student:** "Z axis, positive Z axis."  **Lecturer:** "When you are applying a Hadamard gate on it, where the vector will go?"  **Student:** "Positive x axis."

H2: PAULI GATE OPERATIONS (X, Y, Z)

👤 **Lecturer:** "Now on positive x axis when you will apply Z gate what will happen? Z, X, Y and Z — these all three are NOT gates. That means they will introduce **180-degree phase difference**. The only difference is with respect to which axis they will rotate."

H3: Pauli Gate Rotation Axes

- **X gate:** Rotates with respect to the **X-axis**
- **Y gate:** Rotates with respect to the **Y-axis**
- **Z gate:** Rotates with respect to the **Z-axis**

This is the only difference between these three Pauli gates — they all perform a 180° rotation, but around different axes.

H1: Course Logistics

👤 **Lecturer:** "Thank you. So we will come back by 12:30."

中文全文翻译

一、布拉-凯特符号深入探讨

区分布拉与凯特

讲师继续讨论布拉向量和凯特向量在符号上的区别。

👤 **学生:** "负的。" 🧑 **讲师:** "在符号上，有区别吗？它以竖线开始，以那个符号结束。它以小于号开始，以竖线结束。所以这被称为布拉符号。这被称为凯特。这被称为布拉符号。"

布拉符号的定义

布拉符号将凯特向量转换为行向量，并且对于其中出现的任何复数，取其复共轭。

- 凯特向量表示为列向量
- 布拉向量表示为行向量

- 布拉是凯特的**共轭转置**

复共轭解释

👤 讲师: "什么是复共轭? 任何复数, 它的复共轭是.....如果复数是 $a + bi$"

对于实数 (虚部为零的情况), 复共轭就是它本身。在这个例子中, 两个数都是实数, 所以没有复数部分。

👤 讲师: "所以这是?" 🧑 学生: "0 和 1。" 🧑 学生: "是的。"

布拉与凯特的矩阵乘法

维度分析

👤 讲师: "现在你能把这个向量和这个向量相乘吗?" 🧑 学生: "能乘吗? 这是 rho 乘以 1, 1。" 🧑 讲师: "你能把这个向量相乘吗? 这意味着这是我们的矩阵? 一个简单的 2×1 矩阵, 这是一个 1×2 矩阵。" 🧑 学生: "是的, 可以。" 🧑 讲师: "所以这是一个 1×2 矩阵, 这是一个 2×1 矩阵。你能把 2×1 矩阵和 1×2 矩阵相乘吗? 可以。因为 1 和 1 是共同的。"

内积 (布拉-凯特)

当你将一个**凯特** (列向量) 与一个**布拉** (行向量) 相乘时, 你会得到一个**标量**。

示例:

$$|\psi\rangle = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \langle\phi| = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$\langle\phi|\psi\rangle = \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \times 1 + 2 \times 2 = 5 \text{ (标量)}$$

👤 讲师: "所以这是凯特点布拉。当你将它们相乘时, 你会得到一个标量。你会得到一个标量。你会得到[结果]。"

在量子计算中的意义

👤 讲师: "在量子计算中, 它也有重要意义。这个数字是什么, 它也会揭示一些东西。无论如何, 我今天的目标是让你们理解什么是凯特符号, 什么是布拉符号。至于它们如何被使用, 我们稍后会全部理解。"

二、可视化门操作

阿达玛门应用

👤 讲师："现在你们只需要可视化，然后我们会运行搜索。你们需要可视化阿达玛应用。有一个零向量，你在它上面应用阿达玛门，那么向量最终会到哪里？"

初始状态分析

- 向量最初位于 **Z 轴** 上，具体是 **正 Z 轴**

👤 讲师："之前向量在哪个轴上？" 🧑 学生："Z 轴，正 Z 轴。" 👤 讲师："当你对它应用阿达玛门时，向量会去哪里？" 🧑 学生："正 X 轴。"

泡利门操作 (X、Y、Z)

👤 讲师："现在在正 X 轴上，当你应用 Z 门时会发生什么？Z、X、Y 和 Z——这三个都是非门。这意味着它们会引入 **180 度相位差**。唯一的区别是它们相对于哪个轴旋转。"

泡利门旋转轴

- **X 门**：相对于 **X 轴** 旋转
- **Y 门**：相对于 **Y 轴** 旋转
- **Z 门**：相对于 **Z 轴** 旋转

这是这三个泡利门之间唯一的区别——它们都执行 180° 旋转，但围绕不同的轴。

三、课程事务

👤 讲师："谢谢。我们 12:30 回来。"

好的，请查收根据您提供的字幕文件和要求处理后的内容。

IV. Z Gate and Rotation Direction

Z GATE ON A POSITIVE X VECTOR

- The lecturer continues the discussion on the Z gate, applying it to a vector currently on the positive X-axis.
- Applying a Z gate to a vector on the positive X-axis will rotate it to the **negative X-axis**.
- This is because the Z gate introduces a 180-degree phase difference, rotating the vector around the Z-axis.

 **Student:** "Now on positive x axis when you will apply z gate what will happen?"

 **Lecturer:** "z, x, y and z these all three are not gate that means they will introduce 180 degree phase difference. The only difference is so with respect to which axis they will rotate. X will rotate with respect to x axis, y will rotate with respect to y axis, z will rotate with respect to z axis. That is the only difference. So where the vector will land up? Minus z. So first you have a zero vector, it will land up here. Then on this vector you are applying z gate that means not you are making a note of it like minus x, minus x. If you will notice this, yeah it's minus x. Okay, can you notice minus x everyone? So that was the vector. When we applied on it another operator that is a z operator, then the vector is... So how this rotation will happen? This is a z axis. With respect to z axis it gets rotated by anti-clockwise 180 degree and then the vector will end up there."

UNDERSTANDING ROTATION DIRECTION: ANTI-CLOCKWISE VS. CLOCKWISE

- In spherical coordinates, **anti-clockwise rotation is considered positive**, and **clockwise rotation is considered negative**.
- A negative angle (e.g., -45 degrees) is equivalent to a large positive angle (e.g., 315 degrees).
- This is because rotating clockwise by 45 degrees is the same as rotating anti-clockwise by 315 degrees ($360 - 45 = 315$).

 **Lecturer:** "So in rotation, anti-clockwise is positive. Anti-clockwise is positive. So if you can notice this angle, I am saying as a positive theta. Okay, negative means clockwise. So in spherical coordinate, anything negative angle is clockwise thing, clockwise. So minus 45 degree is what? Positive angle minus 45 degree is what? So minus 45 will be somewhere here. It is minus 45 degree. So in positive, if you have to write this angle, what you will write? How much you have to rotate to do the same operation which minus 45 degree will do? How much you have to rotate? 360 minus 45. 360 minus 45. So what is 360 minus 45? So you have to rotate 90 degree, 90 degree, 90 degree, and then 45 degree more. So 90, 180, 270, 270 plus 45 is 315 degree. So that's what. To minus 45 degree is equivalent to positive 315."

 **Student:** "Yeah."

 **Lecturer:** "So they both are same. So this minus 45 degree phase difference and positive 315 degree phase difference, there is no difference in it. It's same. Same way. So this is anti-clockwise or means my intention is that whether this vector will go this way or this way. Anti-clockwise. It will go this way anticlockwise."

V. Gate Combinations and Preview

UPCOMING GATE OPERATIONS

- The lecturer introduces the concept of combining quantum gates in sequence.
- A dedicated class will be held later to discuss all gates in detail.
- For now, a preview of some key gate combinations is provided:
 - **Hadamard and X gate:** Already discussed.
 - **Hadamard and Z gate:** Already discussed.
 - **Hadamard and S gate:** A new gate, the S gate, is introduced.

INTERACTIVE CHECK: HADAMARD + X GATE

- The lecturer checks the class's understanding of a specific sequence: applying a Hadamard gate followed by an X gate.

 **Lecturer:** "Now, next, there are a lot of gate combinations. First is Hadamard and X we have already seen. Hadamard and X we will see. Hadamard and Z we have already seen. Hadamard and S gate. There is another gate coming up, S gate. See, all these gates exclusively I will take a class where we will just discuss gates. But as and when it is coming, I'm just giving you a hint that what these gates will do. So now the first operation is... First operation, if you can notice all of you please. So this is only Hadamard. Everyone will agree on this. We have already done. X gate we have done. Hadamard gate and X gate. Yeah. Here I want to take your, like I want to understand your understanding. Hadamard gate and X gate. We have applied Hadamard gate followed by X gate. What will happen? Hadamard gate followed by X gate, what will happen? Yeah, still everyone agrees with his statement that it will stay there."

中文全文翻译

四、Z门与旋转方向

对正X轴向量应用Z门

- 讲师继续讨论Z门，将其应用于当前位于正X轴上的向量。
- 对正X轴上的向量应用Z门，会将其旋转到**负X轴**。
- 这是因为Z门引入了180度的相位差，使向量绕Z轴旋转。

🧑 学生：“现在，当你在正X轴上应用Z门时，会发生什么？”

🧑 讲师：“Z、X、Y和Z，这三个都是非门，意味着它们会引入180度的相位差。唯一的区别是它们相对于哪个轴旋转。X将相对于X轴旋转，Y将相对于Y轴旋转，Z将相对于Z轴旋转。这就是唯一的区别。那么向量最终会落在哪里？负Z。所以首先你有一个零向量，它会落在这里。然后在这个向量上你应用Z门，这意味着你把它记作负X，负X。如果你注意到这个，是的，它是负X。好的，大家都能注意到是负X吗？所以那就是那个向量。当我们对它应用另一个算子，即Z算子时，那么向量就是……那么这个旋转是如何发生的？这是Z轴。相对于Z轴，它逆时针旋转180度，然后向量最终会到达那里。”

理解旋转方向：逆时针 vs. 顺时针

- 在球坐标系中，**逆时针旋转被认为是正的，而顺时针旋转被认为是负的**。
- 一个负角度（例如，-45度）等价于一个大的正角度（例如，315度）。
- 这是因为顺时针旋转45度与逆时针旋转315度（ $360 - 45 = 315$ ）是相同的。

🧑 讲师：“所以在旋转中，逆时针是正的。逆时针是正的。所以如果你能注意到这个角度，我称之为正 θ 。好的，负的意味着顺时针。所以在球坐标系中，任何负角度都是顺时针方向。那么负45度是什么？正角度负45度是什么？所以负45度会在这里某处。它是负45度。所以如果用正角度，你要怎么写这个角度？要执行与负45度相同的操作，你需要旋转多少？你需要旋转多少？360减去45。360减去45。那么360减去45是多少？所以你需要旋转90度，90度，90度，然后再多45度。所以90，180，270，270加45是315度。就是这样。所以负45度等价于正315度。”

🧑 学生：“是的。”

🧑 讲师：“所以它们是一样的。所以这个负45度的相位差和正315度的相位差，没有任何区别。是一样的。同样的道理。所以这是逆时针，或者说我的意思是这个向量会走这条路还是那条路。逆时针。它会逆时针走这条路。”

五、门组合与预览

即将介绍的门操作

- 讲师介绍了按顺序组合量子门的概念。
- 稍后将专门安排一节课详细讨论所有门。
- 现在，先预览一些关键的门组合：
 - **Hadamard门和X门**：已经讨论过。
 - **Hadamard门和Z门**：已经讨论过。
 - **Hadamard门和S门**：引入了一个新门，S门。

互动检查：HADAMARD门 + X门

- 讲师检查全班对一个特定序列的理解：先应用Hadamard门，再应用X门。

 **讲师**：“接下来，有很多门组合。首先是Hadamard和X，我们已经看过了。Hadamard和X我们会看到。Hadamard和Z我们已经看过了。Hadamard和S门。还有另一个门即将出现，S门。看，所有这些门，我会专门上一节课来讨论。但是随着它们的出现，我只是给你们一个提示，这些门会做什么。所以现在第一个操作是.....第一个操作，如果你们都能注意到的话，请看。所以这只是Hadamard门。大家都会同意这一点。我们已经做过了。X门我们也做过了。Hadamard门和X门。是的。这里我想听听你们的，我想了解你们的理解。Hadamard门和X门。我们应用了Hadamard门，然后是X门。会发生什么？Hadamard门后接X门，会发生什么？是的，仍然每个人都同意他的说法，即它会停留在那里。”

Quantum Computing Lecture - Gate Operations Analysis

H2: Gate Combinations and Their Effects

H3: HADAMARD GATE FOLLOWED BY X GATE

The lecturer continues the discussion about applying a **Hadamard gate** followed by an **X gate** on a quantum state.

 **Lecturer**: "Everyone will agree on this. We have already done. X gate we have done. Hadamard gate and X gate. Yeah. Here I want to take your, like I want to understand your understanding. Hadamard gate and X gate. We have applied Hadamard gate followed by X gate. What will happen? Hadamard gate followed by X gate. What will happen?"

 **Student**: [Student provides an answer]

 **Lecturer**: "Yeah, still everyone agrees with his statement that it will stay there. It will stay there. Nothing will change."

H3: EXPLANATION OF THE RESULT

The lecturer explains why the state remains unchanged:

- This is the gate combination: **Hadamard gate** followed by **X gate**
- The result shows the state is **still there** - nothing happened
- **Why?** Because the **X gate** is supposed to perform a **NOT operation**, meaning a **180-degree rotation**
- However, with respect to the **X-axis**, the X gate's axis is itself
- Any vector that lies on the X-axis, when you try to rotate it, will just rotate in place
- The vector will still be there on the axis - nothing will change

H3: Z GATE ON ZERO VECTOR

The lecturer extends the discussion to the **Z gate**:

 **Lecturer:** "If there is a 0 vector and I am applying a Z gate, what will happen? 0 vector and I am applying a Z gate. What will happen? 0 vector and I am applying a Z gate on it. Z gate is supposed to rotate this vector with respect to Z axis. Now any vector which is on top of it, nothing will happen to it, so it will stay."

Key observation:

- Applying a **Z gate** on a $|0\rangle$ state will **not** turn it into $|1\rangle$
- If you apply **X gate** or **Y gate**, $|0\rangle$ will turn into $|1\rangle$
- But on the $|0\rangle$ vector, applying a **Z gate** results in **nothing happening** - it stays there

H3: X GATE ON ZERO AND ONE VECTORS

The lecturer demonstrates the **X gate** behavior:

- If we apply **X gate** on $|0\rangle$, it becomes $|1\rangle$
- If we apply **X gate** on $|1\rangle$, it becomes $|0\rangle$ again

 **Lecturer:** "If I apply X gate on 0, then it is going to be 1. It will be 1. If I apply X gate on 1, what will happen? 1 will convert it into 0. It will once again come back to the 0 state."

H3: REVERSIBILITY OF QUANTUM GATES

The lecturer introduces a crucial concept - **reversibility** in quantum computing:

 **Lecturer:** "So having said this, he has raised a good point and it occurs to my mind as well. So what happens in quantum computing? All these gates - these gates are so fascinating - these are **reversible in nature**."

What does reversibility mean?

- If you apply the **same gate twice**, you will return to your **original state**
- This applies to **any gate** in quantum computing

Examples of reversibility:

- **X gate applied twice:** The vector goes once, then returns - **X and X nullify each other**
- **Hadamard gate applied twice:** Apply once for **superposition**, apply again to return to $|0\rangle$ state

 **Lecturer:** "X is easy to understand, but the point is all other gates in quantum computing also behave this way. So if you will apply Hadamard gate twice, you will back to 0 state. You will back to 0 state. So you apply one time Hadamard - superposition. On superposition state, once again apply Hadamard gate - back to 0."

H3: CONNECTION TO CLASSICAL COMPUTING

The lecturer connects this to classical computing:

- This **reversible nature** allows us to use the **NOT gate** from conventional computing as part of quantum gates
- In the conventional world, only the **NOT gate** is **reversible in nature**

中文全文翻译

量子计算讲座 - 门操作分析

门组合及其效果

哈达玛门后接X门

讲师继续讨论对量子态应用**哈达玛门**后接**X门**的情况。

👤 讲师："大家都会同意这一点。我们已经做过了。X门我们做过了。哈达玛门和X门。是的。这里我想听听你们的，我想了解你们的理解。哈达玛门和X门。我们应用了哈达玛门，然后是X门。会发生什么？哈达玛门后接X门。会发生什么？"

👤 学生：[学生给出了一个答案]

👤 讲师："是的，仍然每个人都同意他的说法，即它会停留在那里。它会停留在那里。什么都不会改变。"

结果解释

讲师解释了为什么状态保持不变：

- 这是门组合：**哈达玛门**后接**X门**
- 结果显示状态**仍然在那里** - 什么也没发生
- **为什么？** 因为**X门**应该执行**非运算**，意味着**180度旋转**
- 然而，相对于**X轴**，X门的轴就是它本身
- 任何位于X轴上的向量，当你试图旋转它时，它只会原地旋转
- 向量仍然会在轴上 - 什么都不会改变

对零向量应用Z门

讲师将讨论扩展到**Z门**：

👤 讲师："如果有一个0向量，我对其应用Z门，会发生什么？0向量，我对其应用Z门。会发生什么？0向量，我对其应用Z门。Z门应该相对于Z轴旋转这个向量。现在任何位于其上的向量，什么都不会发生，所以它会停留在那里。"

关键观察：

- 对 $|0\rangle$ 态应用**Z门****不会将其变成 $|1\rangle$**
- 如果你应用**X门**或**Y门**， $|0\rangle$ 会变成 $|1\rangle$
- 但在 $|0\rangle$ 向量上，应用**Z门**的结果是**什么都不会发生** - 它停留在那里

对零向量和一向量应用X门

讲师演示了**X门**的行为：

- 如果我们对 $|0\rangle$ 应用X门，它会变成 $|1\rangle$
- 如果我们对 $|1\rangle$ 应用X门，它会再次变回 $|0\rangle$

👤 讲师："如果我对0应用X门，那么它会变成1。它会是1。如果我对1应用X门，会发生什么？1会变成0。它会再次回到0态。"

量子门的可逆性

讲师引入了一个关键概念——量子计算中的可逆性：

👤 讲师："说到这里，他提出了一个很好的观点，我也想到了这一点。那么在量子计算中会发生什么？所有这些门——这些门太迷人了——它们在本质上是可逆的。"

可逆性意味着什么？

- 如果你应用**同一个门两次**，你将回到你的**原始状态**
- 这适用于量子计算中的**任何门**

可逆性的例子：

- **X门应用两次**：向量去一次，然后返回——**X和X相互抵消**
- **哈达玛门应用两次**：应用一次得到**叠加态**，再次应用回到 $|0\rangle$ 态

👤 讲师："X很容易理解，但关键是量子计算中所有其他门也这样表现。所以如果你应用哈达玛门两次，你会回到0态。你会回到0态。所以你应用一次哈达玛门——叠加态。在叠加态上，再次应用哈达玛门——回到0。"

与经典计算的联系

讲师将此与经典计算联系起来：

- 这种**可逆性质**允许我们将传统计算中的非门作为量子门的一部分使用
- 在传统世界中，只有**非门**在本质上是可逆的

Quantum Computing Lecture - Continued

Reversibility in Quantum Gates

CORE CONCEPT OF REVERSIBILITY

The lecturer elaborates on the **reversibility** property of quantum gates, which is a fundamental characteristic distinguishing them from classical gates.

- Applying a **Hadamard gate** twice returns the qubit to its original state ($|0\rangle \rightarrow$ superposition $\rightarrow |0\rangle$)
- This **reversible nature** allows us to incorporate the classical **NOT gate** into quantum gates
- In the classical world, only the **NOT gate** is inherently **reversible**

Definition of Reversibility:

- Applying the same gate twice returns you to your input state
- From the output, you can determine what the input was

EXAMPLE: NOT GATE REVERSIBILITY

 **Lecturer:** "There is a NOT gate and you are getting an output 0. What was the possible input? You are getting 0 on NOT gate. What was the possible input on that NOT gate? 1. So just by looking at the output, I can infer back, I can go reverse. I can reverse it and find out what was the input."

EXAMPLE: AND GATE IRREVERSIBILITY

 **Lecturer:** "If I will give you an AND gate. So AND gate has 2 inputs. And there is one output to AND gate. So if I will give you an output 0, so can you tell me what were the input combinations on two inputs? You can't tell. So it is not reversible."

KEY CONCLUSION

- **All quantum gates are reversible in nature**
 - This **reversibility** is critical for both:
 - Conceptual understanding
 - Hardware implementation
 - All gates used in quantum computing must be **reversible**
-

Special Quantum Gates: S, T, and Z Gates

ROTATION GATES AROUND Z-AXIS

The lecturer introduces three special gates that all rotate around the **Z-axis** but with different **phase angles**:

Gate	Rotation Axis	Rotation Degree	Phase
Z gate	Z-axis	180°	π
S gate	Z-axis	90°	$\pi/2$
T gate	Z-axis	45°	$\pi/4$

 **Lecturer:** "Their rotation axis is same. So, they all S, T and Z they will rotate with respect to Z axis but the degree is different, the phase is different. 180 degree is for Z gate, S 90 degree, T 45 degree. That is the only difference."

GATE COMBINATIONS

- **S followed by T** creates a specific combination
- With respect to the positive 90° **S gate**, if you try the **T gate** instead, it will be somewhere at 45°

SUMMARY OF GATES COVERED

The lecturer provides a comprehensive overview of all gates discussed so far:

- **Hadamard gate (H)**
- **X gate**
- **Y gate**
- **Z gate**
- **S gate**
- **T gate**
- **CX gate** (Controlled-X / CNOT)
- **CCX gate** (Toffoli gate)
- **CCCX gate** (and higher-order controlled gates)

👤 **Lecturer:** "Based on CX gate, like the way we design CX gate, we can design CCX gate, we can design CCCX gate, we can add as many conditions..."

中文全文翻译

量子计算讲座（续）

量子门的可逆性

可逆性的核心概念

讲师详细阐述了量子门的**可逆性**属性，这是区分量子门与经典门的基本特征。

- 对量子比特应用两次**哈达玛门**会使其回到初始状态 ($|0\rangle \rightarrow \text{叠加态} \rightarrow |0\rangle$)
- 这种**可逆性质**允许我们将经典的**非门**纳入量子门中
- 在经典世界中，只有**非门**本质上是**可逆的**

可逆性的定义：

- 对同一个门应用两次，你会回到输入状态
- 从输出结果，你可以判断输入是什么

示例：非门的可逆性

👤 **讲师:** "有一个非门，你得到了输出0。可能的输入是什么？你在非门上得到了0。那个非门上可能的输入是什么？是1。所以仅仅通过观察输出，我就能反向推断，我可以逆向推导。我可以逆向推导并找出输入是什么。"

示例：与门的不可逆性

👤 **讲师:** "如果我给你一个与门。与门有两个输入。与门有一个输出。如果我给你输出0，你能告诉我两个输入上的输入组合是什么吗？你无法判断。所以它是不可逆的。"

关键结论

- 所有量子门本质上都是可逆的

- 这种可逆性对于以下两方面都至关重要：
 - 概念理解
 - 硬件实现
- 量子计算中使用的所有门都必须是可逆的

特殊量子门：S门、T门和Z门

绕Z轴的旋转门

讲师介绍了三种特殊门，它们都绕Z轴旋转，但具有不同的相位角度：

门	旋转轴	旋转角度	相位
Z门	Z轴	180°	π
S门	Z轴	90°	$\pi/2$
T门	Z轴	45°	$\pi/4$

👤 讲师："它们的旋转轴是相同的。所以S、T和Z都绕Z轴旋转，但角度不同，相位不同。Z门是180度，S门是90度，T门是45度。这是唯一的区别。"

门的组合

- S门后接T门会形成特定的组合
- 相对于正90度的S门，如果你改用T门，它会在45度附近

已讲解门的总结

讲师全面回顾了到目前为止讨论的所有门：

- 哈达玛门 (H)
- X门
- Y门
- Z门
- S门
- T门
- CX门 (受控X门/CNOT门)

- **CCX**门 (托福利门)
- **CCCX**门 (以及更高阶的受控门)

 **讲师:** "基于CX门, 就像我们设计CX门的方式一样, 我们可以设计CCX门, 我们可以设计CCCX门, 我们可以添加任意多的条件....."

好的, 这是根据您的要求, 对提供的字幕文本进行处理后的结果。

QUANTUM COMPUTING LECTURE - GATE REVIEW, EXERCISES & STATE REPRESENTATION

1. Review of Quantum Gates (English Only)

The lecturer reviews the fundamental quantum gates that have been covered, emphasizing that these form the toolkit for designing algorithms.

- **Single-Qubit Gates:**
 - **Hadamard Gate (H)**
 - **X Gate**
 - **Y Gate**
 - **Z Gate**
 - **S Gate**
 - **T Gate**
- **Multi-Qubit & Conditional Gates:**
 - **CX Gate (Controlled-X / CNOT Gate)**
 - **CCX Gate (Toffoli Gate)**
 - **CCCX Gate** (and higher-order controlled gates)
 - **Swap Gate**
 - **C-Swap Gate (Fredkin Gate)**

 **Lecturer:** "Based on the CX gate, like the way we design the CX gate, we can design the CCX gate, we can design the CCCX gate, we can add as many conditions as we want. It is still an X gate, but with two conditions. Until and unless those two conditions get satisfied, then only the X gate will work. So that is the CCX gate."

 **Lecturer:** "Likewise, there is a Swap gate. So there are two qubits and you need to swap them on a condition. Once the condition gets satisfied, then swap the qubits. So that

is called the C-Swap gate. So these, more or less, are the only gates. Using those gates, we have to design some algorithm."

2. Course Exercises & Submission Instructions (English Only)

The lecturer outlines the expected tasks for the students, including a specific exercise on complex numbers and the completion of a Jupyter notebook.

- **Exercise 1: Complex Number Visualization**

- Take two complex numbers.
- Visualize them.
- Perform addition and subtraction.
- Convert from Cartesian coordinates to polar coordinates.
- Convert from polar coordinates to Cartesian coordinates.

- **Exercise 2: Notebook Completion**

- Complete all the reflections/exercises that are part of the current Jupyter notebook (`knot pool`).
- If you have not submitted the previous notebook, you must submit that as well.

- **Submission Method:**

- Maintain a repository on **GitHub** for all your work.
- Share the link to the repository with the lecturer.

 **Student:** "Do we submit it in our Teams?"

 **Lecturer:** "Yeah, on... like on GitHub you can maintain the repository. Whatever the work you are doing, and you can share me just the link."

3. Quantum State Representation (English Only)

The lecturer transitions to a core concept: how to mathematically represent a quantum state.

- **The Qubit State:** A quantum qubit is a combination of the $|0\rangle$ and $|1\rangle$ states.
- **Dirac (Bra-Ket) Notation:** The quantum state is represented using the **ket notation**.
- **Probability Amplitudes:**
 - The state is written as: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
 - α and β are **probability amplitudes**.
 - The square of these amplitudes gives the probability of measuring the corresponding state.
 - $|\alpha|^2$ is the probability of measuring $|0\rangle$.

- $|\beta|^2$ is the probability of measuring $|1\rangle$.
- **Nature of Amplitudes:** α and β can be real numbers or **complex numbers**.
- **Superposition:** This combination of states with probability amplitudes is how superposition is mathematically represented.
- **Base State Vectors:**
 - The $|0\rangle$ state can be represented as a vector/matrix: $[1, 0]^T$
 - The $|1\rangle$ state can be represented as a vector/matrix: $[0, 1]^T$

 **Lecturer:** "So, a quantum qubit is a combination of zero and one. And how we represent it? What is a quantum state? So a quantum state we represent something like this... So this is the quantum state in the ket notation. So this is the qubit state, and this qubit is a combination of zero state and one state. These are two primitive states."

 **Lecturer:** "So, alpha and beta will determine the probability. So these are the probabilistic amplitudes. So these are the probability amplitudes. These amplitudes squared is the probability. So, alpha squared is the probability of? ... Zero. And beta squared is the probability of one. That's how the superpositioning happens. Now, these alpha and beta could be real or a complex number."

中文全文翻译

1. 量子门回顾

讲师回顾了已涵盖的基本量子门，并强调这些门构成了设计算法的工具包。

- **单量子比特门:**
 - 哈达玛门 (H)
 - X门
 - Y门
 - Z门
 - S门
 - T门
- **多量子比特门与条件门:**
 - CX门 (受控X门 / CNOT门)
 - CCX门 (托福利门)
 - CCCX门 (以及更高阶的受控门)
 - 交换门
 - 受控交换门 (弗雷德金门)

👤 讲师：“基于CX门，就像我们设计CX门的方式一样，我们可以设计CCX门，我们可以设计CCCX门，我们可以添加任意多的条件。它仍然是一个X门，但带有两个条件。只有当这两个条件都满足时，X门才会工作。这就是CCX门。”

👤 讲师：“同样地，还有一个交换门。假设有两个量子比特，你需要在一个条件下交换它们。一旦条件满足，就交换这两个量子比特。这被称为受控交换门。所以，这些或多或少就是所有的门了。我们必须使用这些门来设计一些算法。”

2. 课程练习与提交说明

讲师概述了学生需要完成的任务，包括一个关于复数的特定练习以及一个Jupyter notebook的完成。

- **练习1：复数可视化**
 - 取两个复数。
 - 将它们可视化。
 - 执行加法和减法运算。
 - 从笛卡尔坐标转换为极坐标。
 - 从极坐标转换为笛卡尔坐标。
- **练习2：完成Notebook**
 - 完成当前Jupyter notebook（knot pool）中的所有练习/反思部分。
 - 如果你还没有提交之前的notebook，也必须一并提交。
- **提交方式：**
 - 在 **GitHub** 上为你所做的所有工作维护一个仓库。
 - 将仓库的链接分享给讲师。

👤 学生：“我们要在Teams里提交吗？”

👤 讲师：“是的，在.....比如在GitHub上你可以维护一个仓库。无论你做了什么工作，都可以把链接分享给我。”

3. 量子态表示

讲师转向一个核心概念：如何用数学方式表示一个量子态。

- **量子比特态：** 一个量子比特是 $|0\rangle$ 和 $|1\rangle$ 状态的组合。
- **狄拉克符号（括号符号）：** 量子态使用 **ket符号** 来表示。
- **概率幅：**
 - 状态写作： $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
 - α 和 β 是 **概率幅**。

- 这些幅度的平方给出了测量到相应状态的概率。
 - $|\alpha|^2$ 是测量到 $|0\rangle$ 的概率。
 - $|\beta|^2$ 是测量到 $|1\rangle$ 的概率。
- 幅度的性质: α 和 β 可以是实数或复数。
- 叠加态: 这种将状态与概率幅相结合的方式, 就是叠加原理在数学上的体现。
- 基态向量:
 - $|0\rangle$ 状态可以表示为一个向量/矩阵: $[1, 0]^T$
 - $|1\rangle$ 状态可以表示为一个向量/矩阵: $[0, 1]^T$

👤 讲师: “所以, 一个量子比特是0和1的组合。我们如何表示它? 什么是量子态? 我们像这样表示一个量子态.....这就是用ket符号表示的量子态。这就是量子比特态, 这个量子比特是零态和壹态的组合。这是两个基本态。”

👤 讲师: “所以, alpha和beta决定了概率。这些是概率幅。这些是概率幅。这些幅度的平方就是概率。所以, alpha的平方是.....的概率?零。而beta的平方是壹的概率。这就是叠加态发生的方式。现在, 这些alpha和beta可以是实数或复数。”

Of course. Here is the processed content based on the provided subtitle transcript, strictly following all five requirements.

1. Zero Information Loss & Structural Reorganization (English Only)

LECTURE: QUANTUM STATE REPRESENTATION & SIMULATOR FUNDAMENTALS

The lecturer continues the discussion on quantum state representation, specifically focusing on the vector (matrix) form of base states and how this translates to a quantum simulator.

Representing Base States as Vectors

- The base state $|0\rangle$ is represented as a column vector: $[1, 0]^T$.
- The base state $|1\rangle$ is represented as a column vector: $[0, 1]^T$.
- In this representation, the numbers 1 and 0 are **probability amplitudes**.
 - For $|0\rangle$, the amplitude for $|0\rangle$ is 1, and the amplitude for $|1\rangle$ is 0.
- A general qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is written as a column vector $[\alpha, \beta]^T$.
- **Key Point:** In ket notation, the state is *always* represented as a column vector.

🧑‍🎓 **Student:** "Yeah." 🧑‍🎓 **Lecturer:** "Like what is 1? What is 0? Amplitude. Probabilistic amplitude. So 1 is the probabilistic amplitude corresponding to 0 vector. 0 is the 1. So if corresponding to 0, you have to write it in a vector form this alpha plus beta. Now what is alpha? Alpha is 1 and what is beta? Beta is 0. So, this so what we are writing here? We are writing here alpha and beta. So, we are writing this here alpha and beta and cat notation is always in column vector. So, column vector represent. So if in next class, if I will just write zero, technically the meaning is this."

Connecting to the Simulator

The lecturer explains that this vector representation is not just theoretical; it is the fundamental way a quantum simulator works internally.

- **Simulator Design:** A simulator does not use abstract symbols. It uses the matrix representation.
- **Initial State:** For a simulator, the vector $[1, 0]$ is the $|0\rangle$ vector. It is the initial state.
- **Multi-Qubit States:** The lecturer poses a question about representing a multi-qubit state, specifically the state $|10\rangle$.

🧑‍🎓 **Student:** "Okay." 🧑‍🎓 **Lecturer:** "I'm preparing you to understand how simulator works. So in simulator, there is nothing like this. So if you have to represent zero vector, so that's how we are representing on our system. So simulator in its design is using this matrix. So 1, 0 for simulator means 0 vector." 🧑‍🎓 **Student:** "Yes. That could be an initial state." 🧑‍🎓 **Lecturer:** "Yeah, that's the initial state. Then, if it is 1, 0 and 1. Same way, I am asking you if I am writing this state. Can you tell me, last time also I discussed, so please help me this time. If this is the vector, in your simulator, if you have to represent it, It's a cat notation. Cat notation means it's a column vector. What will be the size of it and what will be the values on top of it?"

Representing the $|10\rangle$ State

The lecturer uses the state $|10\rangle$ as an example to demonstrate how to construct a multi-qubit state vector.

- **Number of Qubits:** The state $|10\rangle$ implies a 2-qubit system.
- **Vector Size:** A 2-qubit system has $2^2 = 4$ possible base states. Therefore, the state vector is a 4×1 column vector.
- **Base State Ordering:** The four base states for a 2-qubit system are ordered as:
 1. $|00\rangle$
 2. $|01\rangle$
 3. $|10\rangle$

4. $|11\rangle$

- **Constructing the Vector:** To represent the state $|10\rangle$, the amplitude for the third base state ($|10\rangle$) is 1, and the amplitudes for all other base states are 0.
- **Resulting Vector:** The column vector for $|10\rangle$ is:

```
[0] (amplitude for |00>)
[0] (amplitude for |01>)
[1] (amplitude for |10>)
[0] (amplitude for |11>)
```

🧑 **Student:** "0 and 10." 🧑 **Lecturer:** "1 0 is the vector." 🧑 **Student:** "Okay." 🧑

Lecturer: "So 1 0 vector means how many qubits are there? 2 qubits. So 2 qubits means how many possible number of states? 4. So the vector size will be? 4. 4 cross 1. So the first, this is for the first location is corresponding to 0, 0. The next location is corresponding to 1, 0. Sorry not for 1, 0. 0, 1. 1, 0. And 1, 1. Now corresponding to 1, 0, will initialize if I have to initialize this vector on my system."

Alternative Notation for Multi-Qubit States

The lecturer presents an alternative, more explicit way to write the same state, which is common in textbooks and papers.

- The state $|10\rangle$ can be written as a sum of all base states with their respective amplitudes:

$$\alpha_1 |00\rangle + \alpha_2 |01\rangle + \alpha_3 |10\rangle + \alpha_4 |11\rangle$$

- For the specific state $|10\rangle$:

- $\alpha_1 = 0$
- $\alpha_2 = 0$
- $\alpha_3 = 1$
- $\alpha_4 = 0$

🧑 **Lecturer:** "Normally, in other words, also for normal understanding, if I'm writing a book, if I'm writing a journal or paper. So how I can represent this state, I can represent this state in this way as well. So if 1, 0 is my state, I can write it. $\alpha_1 0 0$ plus $\alpha_2 0 1$ plus $\alpha_3 1 0$ plus $\alpha_4 1 1$. Now what is α_1 ? α_1 is?"

2. Full Chinese Translation (中文全文翻译)

讲座：量子态表示与模拟器基础

讲师继续讨论量子态的表示，特别是基态的向量（矩阵）形式，以及这如何转化为量子模拟器。

将基态表示为向量

- 基态 $|0\rangle$ 被表示为一个列向量： $[1, 0]$ 。
- 基态 $|1\rangle$ 被表示为一个列向量： $[0, 1]$ 。
- 在这种表示中，数字 1 和 0 是概率幅。
 - 对于 $|0\rangle$ ， $|0\rangle$ 的幅值是 1， $|1\rangle$ 的幅值是 0。
- 一个通用的量子比特态 $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ 被写成一个列向量 $[\alpha, \beta]$ 。
- **关键点：** 在狄拉克符号中，状态总是被表示为一个列向量。

👤 学生：“是的。” 🗣️ 讲师：“比如什么是1？什么是0？幅值。概率幅。所以1是对应于0向量的概率幅。0是1。所以如果对应于0，你必须用向量形式写出这个alpha加beta。现在什么是alpha？alpha是1，什么是beta？beta是0。所以，我们在这里写的是什麼？我们在这里写的是alpha和beta。所以，我们在这里写alpha和beta，而ket符号总是列向量。所以，列向量表示。所以如果在下节课，我只写零，从技术上讲，意思就是这个。”

连接到模拟器

讲师解释说，这种向量表示不仅仅是理论上的；它是量子模拟器内部工作的基本方式。

- **模拟器设计：** 模拟器不使用抽象符号。它使用矩阵表示。
- **初始状态：** 对于模拟器，向量 $[1, 0]$ 就是 $|0\rangle$ 向量。它是初始状态。
- **多量子比特态：** 讲师提出了一个关于表示多量子比特态的问题，特别是状态 $|10\rangle$ 。

👤 学生：“好的。” 🗣️ 讲师：“我正在为你们理解模拟器如何工作做准备。所以在模拟器中，没有像这样的东西。所以如果你必须表示零向量，这就是我们在系统上表示它的方式。所以模拟器在其设计中使用了这个矩阵。所以对于模拟器来说，1, 0 意味着0向量。” 🤖
👤 学生：“是的。那可以是一个初始状态。” 🗣️ 讲师：“是的，那是初始状态。然后，如果它是1, 0 和 1。同样地，我问你，如果我写这个状态。你能告诉我吗，上次我也讨论过，所以这次请帮我一下。如果这是向量，在你的模拟器中，如果你必须表示它，它是ket符号。ket符号意味着它是一个列向量。它的大小是多少，它上面的值是什麼？”

表示 $|10\rangle$ 态

讲师使用状态 $|10\rangle$ 作为示例，演示如何构建一个多量子比特态向量。

- **量子比特数：** 状态 $|10\rangle$ 暗示了一个2量子比特系统。

- **向量大小：** 一个2量子比特系统有 $2^2 = 4$ 个可能的基态。因此，状态向量是一个 4×1 的列向量。
- **基态排序：** 一个2量子比特系统的四个基态排序如下：
 1. $|00\rangle$
 2. $|01\rangle$
 3. $|10\rangle$
 4. $|11\rangle$
- **构建向量：** 为了表示状态 $|10\rangle$ ，第三个基态 ($|10\rangle$) 的幅值为 1，所有其他基态的幅值为 0。
- **结果向量：** $|10\rangle$ 的列向量是：

```
[0]    (对应于 |00> 的幅值)
[0]    (对应于 |01> 的幅值)
[1]    (对应于 |10> 的幅值)
[0]    (对应于 |11> 的幅值)
```

👤 学生：“0 和 10。” 🧑🏫 讲师：“1 0 是向量。” 👤 学生：“好的。” 🧑🏫 讲师：“所以 1 0 向量意味着有多少个量子比特？2个量子比特。所以2个量子比特意味着有多少个可能的状态？4个。所以向量大小是？4。4 乘 1。所以第一个，这个第一个位置对应于 0, 0。下一个位置对应于 1, 0。抱歉不是 1, 0。是 0, 1。然后是 1, 0。然后是 1, 1。现在对应于 1, 0，如果我必须在我的系统上初始化这个向量，就会初始化它。”

多量子比特态的替代记法

讲师提出了另一种更显式的写法来表示同一个状态，这在教科书和论文中很常见。

- 状态 $|10\rangle$ 可以写成所有基态与其各自幅值的和：

$$\alpha_1 |00\rangle + \alpha_2 |01\rangle + \alpha_3 |10\rangle + \alpha_4 |11\rangle$$

- 对于特定的状态 $|10\rangle$ ：

- $\alpha_1 = 0$
- $\alpha_2 = 0$
- $\alpha_3 = 1$
- $\alpha_4 = 0$

🧑🏫 讲师：“通常，换句话说，为了正常理解，如果我在写一本书，如果我在写一本期刊或论文。那么我如何表示这个状态，我也可以用这种方式表示这个状态。所以如果 1, 0 是我的状态，我可以这样写。alpha 1 0 0 加上 alpha 2 0 1 加上 alpha 3 1 0 加上 alpha 4 1 1。现在什么是 alpha 1？alpha 1 是？”

Quantum Computing Lecture - Level 9 Master of Software Engineering

1. State Vector Representation in Literature vs. Simulation

NOTATION IN ACADEMIC CONTEXT

 **Lecturer:** "If I'm writing a book, if I'm writing a journal or paper, how can I represent this state? I can represent this state in this way as well."

The lecturer explains that when writing for publication or discussion, a different notation is used compared to what is needed for simulation.

For a 2-qubit system with state $|1,0\rangle$:

- The state can be written as:
 $\alpha_1 |00\rangle + \alpha_2 |01\rangle + \alpha_3 |10\rangle + \alpha_4 |11\rangle$

Where the coefficients are:

- $\alpha_1 = 0$ (amplitude for $|00\rangle$)
- $\alpha_2 = 0$ (amplitude for $|01\rangle$)
- $\alpha_3 = 1$ (amplitude for $|10\rangle$)
- $\alpha_4 = 0$ (amplitude for $|11\rangle$)

 **Lecturer:** "So that's what the state vector is. On simulator, I need this. If I'm just having a discussion with someone, I'm writing a book, or I'm reading any book, you will find this kind of notation."

WHY VECTOR/MATRIX NOTATION IS NECESSARY FOR SIMULATION

 **Lecturer:** "Because we are simulating on our machine, we have to represent these states, so then we will use these vector formations, matrix formations. What I am preparing you for, I am preparing you to understand how simulator works."

2. How a Quantum Simulator Works

THE SIMULATION PROCESS

The lecturer walks through what happens when simulating a quantum circuit:

1. **Initial State:** A qubit starts in the initial zero state $|0\rangle$ represented as vector $[1, 0]$
2. **Apply Hadamard Gate:** The Hadamard operator is applied
3. **Apply S Gate:** The S gate is applied
4. **Result:** A final state vector is obtained

 **Lecturer:** "So what is the initial state? Initial state is 1 0. And let's assume that you come up with some other state. So this is your alpha 1 and alpha 2, final state."

MATRIX MANIPULATION AS THE CORE OPERATION

 **Lecturer:** "What exactly will simulator do? Simulator knows the Hadamard operator. Corresponding to the Hadamard operator, there is a matrix. When you have to apply a Hadamard operator, there is a matrix associated with the Hadamard gate. Likewise for S gate, when you will apply, there is a matrix associated with it. There is a matrix which is representing my initial state."

The Simulator's Process:

- The simulator performs **matrix manipulations** (multiplication, addition, etc.)
- Starting from the initial state, it applies each gate's matrix sequentially
- Each gate has a corresponding matrix representation

 **Lecturer:** "So your simulator is only going to do the matrix manipulation. So initial state, apply the Hadamard gate. Corresponding to Hadamard gate, there is some kind of a matrix. Corresponding to X-gate, there is kind of a matrix. Then simulator will just do the manipulation of those matrices. So starting from this state to that next state, that next state."

3. The Scalability Challenge

MATRIX SIZE GROWTH WITH QUBIT COUNT

 **Lecturer:** "Now the problem is when you will expand your qubit size, 5 qubit, 10 qubit, more than that, your matrix size will also increase."

Key Insight:

- As the number of qubits increases, the matrix size grows exponentially
- Matrix multiplication and manipulation become computationally hard problems
- This is why simulation becomes challenging for larger quantum systems

👤 **Lecturer:** "When the matrix size will increase, you know the matrix multiplication or matrix manipulation is not so easy. It also comes under the hard problem when the matrix size will increase."

中文全文翻译

1. 文献与模拟中的状态向量表示

学术语境中的符号表示

👤 **讲师:** "如果我在写一本书，如果我在写一本期刊或论文，我该如何表示这个状态？我也可以用这种方式表示这个状态。"

讲师解释说，在撰写出版物或进行学术讨论时，使用的符号表示法与模拟所需的表示法不同。

对于2量子比特系统，状态为 $|1,0\rangle$:

- 该状态可以写成:
$$\alpha_1 |00\rangle + \alpha_2 |01\rangle + \alpha_3 |10\rangle + \alpha_4 |11\rangle$$

其中系数为:

- $\alpha_1 = 0$ ($|00\rangle$ 的振幅)
- $\alpha_2 = 0$ ($|01\rangle$ 的振幅)
- $\alpha_3 = 1$ ($|10\rangle$ 的振幅)
- $\alpha_4 = 0$ ($|11\rangle$ 的振幅)

👤 **讲师:** "这就是状态向量。在模拟器上，我需要这个。如果我只是在和别人讨论，或者我在写书、读书，你会看到这种符号表示。"

为什么模拟需要向量/矩阵表示

👤 讲师："因为我们在机器上进行模拟，我们必须表示这些状态，所以我们会使用这些向量形式、矩阵形式。我给你们做的准备，是为了让你们理解模拟器是如何工作的。"

2. 量子模拟器的工作原理

模拟过程

讲师逐步讲解模拟量子电路时发生的过程：

1. 初始状态：量子比特从初始零态 $|0\rangle$ 开始，表示为向量 $[1, 0]$
2. 应用Hadamard门：应用Hadamard算子
3. 应用S门：应用S门
4. 结果：得到最终状态向量

👤 讲师："那么初始状态是什么？初始状态是1 0。假设你得到了另一个状态。这就是你的 α_1 和 α_2 ，最终状态。"

矩阵操作作为核心运算

👤 讲师："模拟器具体会做什么？模拟器知道Hadamard算子。对应于Hadamard算子，有一个矩阵。当你需要应用Hadamard算子时，有一个与Hadamard门相关联的矩阵。同样地，对于S门，当你应用它时，也有一个与之关联的矩阵。还有一个表示我初始状态的矩阵。"

模拟器的过程：

- 模拟器执行**矩阵操作**（乘法、加法等）
- 从初始状态开始，依次应用每个门的矩阵
- 每个门都有对应的矩阵表示

👤 讲师："所以你的模拟器只会做矩阵操作。从初始状态开始，应用Hadamard门。对应于Hadamard门，有某种矩阵。对应于X门，也有某种矩阵。然后模拟器就会对这些矩阵进行操作。从当前状态到下一个状态，再到下一个状态。"

3. 可扩展性挑战

量子比特数量增加导致的矩阵规模增长

👤 讲师: "现在的问题是, 当你扩大量子比特规模时, 5个量子比特、10个量子比特, 甚至更多, 你的矩阵规模也会随之增大。"

关键洞察:

- 随着量子比特数量的增加, 矩阵规模呈指数级增长
- 矩阵乘法和操作在计算上变得困难
- 这就是为什么对大型量子系统进行模拟变得具有挑战性

👤 讲师: "当矩阵规模增大时, 你知道矩阵乘法或矩阵操作并不容易。当矩阵规模增大时, 这也属于困难问题的范畴。"

Quantum Computing Lecture - Matrix Manipulation & Simulator Limitations

The Scaling Problem of Quantum Simulators

MATRIX SIZE EXPLOSION WITH QUBIT COUNT

As the number of qubits increases, the matrix representation grows exponentially:

- **5 qubits** → manageable matrix size
- **10 qubits** → significantly larger matrix
- **30 qubits** → matrix becomes $2^{30} \times 2^{30}$ (approximately 1 billion × 1 billion)

👤 **Lecturer:** "Now the problem is when you will expand your qubit size, 5 qubit, 10 qubit, more than that, your matrix size will also increase."

COMPUTATIONAL CHALLENGES

The fundamental difficulty lies in matrix operations:

- **Matrix multiplication** and **matrix manipulation** become computationally expensive
- As matrix size grows, these operations fall into the category of **hard problems**
- Beyond a certain point, conventional machines cannot solve these problems

👤 **Lecturer:** "When the matrix size will increase, you know the matrix multiplication or matrix manipulation is not so easy. It is also comes under the hard problem when the

matrix size will increase."

Simulator Design & Optimization

THE OPTIMIZATION GAP

Building a custom simulator is possible, but faces significant challenges:

- **Without proper optimization**, the simulator will not function effectively
- Professional simulators (e.g., **Qiskit**) implement **highly optimized code**
- Optimization enables fast execution despite large matrix sizes

 **Lecturer:** "So you can also design your simulator, but the only problem is that your simulator will not work. Because you will not do the optimization."

CORE PRINCIPLE OF SIMULATORS

All quantum simulators share the same fundamental approach:

1. **Represent states** through matrix formation
2. **Represent operators** through matrix formation
3. **Perform matrix operations** (multiplication, manipulation)

 **Lecturer:** "So it is all about representing your states through matrix formation, representing your operators through matrix formation and then do these matrix operations."

THE 30-QUBIT BARRIER

A concrete example illustrates the exponential challenge:

- **30 qubits** → matrix of size $2^{30} \times 2^{30}$
- This matrix is **impossible** to manipulate on conventional hardware
- Even simulators will eventually fail beyond this point

 **Lecturer:** "Just imagine 30 qubits. With 30 qubits, there may be a formation of a matrix which may be almost 2 raised to the power 30 cross 2 raised to the power 30. Just imagine the size of that matrix."

PRACTICAL LIMITATIONS IN DEMONSTRATIONS

Due to these constraints, lecture demonstrations are limited:

- Examples will not exceed **3-5 qubits**
- Larger simulations would take **excessive time** on available systems

 **Lecturer:** "So that's why even in my example also I will not go beyond 3 to 4, 5, because then it will take ages for my system to simulate them."

Course Logistics & Next Steps

PREPARATION FOR NEXT CLASS

Required preparation:

- Complete all previously assigned tasks
- Review **matrix manipulation** concepts:
 - Matrix multiplication
 - Matrix division
 - Understanding what matrices represent
 - Operators applicable to matrices

Content availability:

- Updated content will be shared on **Teams**
- **Blackboard** content will remain largely the same
- Additional supplementary material will be provided

 **Lecturer:** "So next class you will start from this notebook and before that whatever I have asked you to do please do it."

STUDENT INTERACTION

 **Student:** "Should we ask him to post or work on our teams?"

 **Lecturer:** "That's also similar."

 **Student:** "I think that will be easier for us to remember."

UPCOMING TOPICS

The next session will cover:

1. **Matrix operations** (multiplication, division)
2. **Gate-based quantum operations**
3. Understanding **matrix representation** of quantum states and operators

 **Lecturer:** "So apart from this, there will be some other stuff which we will do through gates. But I want you to actually understand what the matrix is, what sort of an operators we can apply on matrix, what we can represent through matrix, that kind of stuff."

中文全文翻译

量子计算课程 - 矩阵操作与模拟器局限性

量子模拟器的规模问题

量子比特数量导致的矩阵规模爆炸

随着量子比特数量的增加，矩阵表示呈指数级增长：

- 5个量子比特 → 可管理的矩阵规模
- 10个量子比特 → 显著增大的矩阵
- 30个量子比特 → 矩阵变为 $2^{30} \times 2^{30}$ (约10亿×10亿)

 **讲师：** "现在的问题是，当你扩大量子比特规模时，5个量子比特、10个量子比特，甚至更多，你的矩阵规模也会随之增大。"

计算挑战

根本困难在于矩阵操作：

- 矩阵乘法和矩阵操作在计算上变得昂贵

- 随着矩阵规模增大，这些操作属于**困难问题**的范畴
- 超过某个临界点后，传统计算机无法解决这些问题

👤 讲师："当矩阵规模增大时，你知道矩阵乘法或矩阵操作并不容易。当矩阵规模增大时，这也属于困难问题的范畴。"

模拟器设计与优化

优化差距

构建自定义模拟器是可能的，但面临重大挑战：

- **没有适当的优化**，模拟器将无法有效运行
- 专业模拟器（如**Qiskit**）实现了**高度优化的代码**
- 优化使得在大矩阵规模下仍能快速执行

👤 讲师："所以你也可以设计自己的模拟器，但唯一的问题是**你的模拟器无法工作**。因为你没有进行优化。"

模拟器的核心原理

所有量子模拟器共享相同的基本方法：

1. **通过矩阵形式表示状态**
2. **通过矩阵形式表示算子**
3. **执行矩阵操作（乘法、操作）**

👤 讲师："所以这一切都归结为**通过矩阵形式表示你的状态**，**通过矩阵形式表示你的算子**，然后进行这些矩阵操作。"

30量子比特的障碍

一个具体例子说明了指数级挑战：

- **30个量子比特** → 矩阵规模为 $2^{30} \times 2^{30}$
- 这个矩阵在传统硬件上**不可能**进行操作
- 超过这个点，即使是模拟器最终也会失效

👤 讲师: "想象一下30个量子比特。30个量子比特可能会形成一个几乎是2的30次方乘以2的30次方的矩阵。想象一下那个矩阵的大小。"

演示中的实际限制

由于这些限制，课堂演示受到限制：

- 示例不会超过**3-5个量子比特**
- 更大的模拟在现有系统上需要**过长时间**

👤 讲师: "所以这就是为什么即使在我的示例中，我也不会超过3到4、5个量子比特，因为那样我的系统需要很长时间来模拟它们。"

课程安排与后续步骤

下节课准备

需要完成的准备工作：

- 完成所有先前布置的任务
- 复习**矩阵操作**概念：
 - 矩阵乘法
 - 矩阵除法
 - 理解矩阵表示的内容
 - 可应用于矩阵的算子

内容获取方式：

- 更新后的内容将在**Teams**上分享
- **Blackboard**上的内容将基本保持不变
- 将提供额外的补充材料

👤 讲师: "所以下节课你们将从这本笔记本开始，在此之前，请完成我要求你们做的所有事情。"

师生互动

👤 学生: "我们应该让他发布到Teams上吗？"

 **讲师：** "那也是一样的。"

 **学生：** "我觉得那样我们更容易记住。"

即将讨论的主题

下节课将涵盖：

1. 矩阵操作（乘法、除法）
2. 基于门的量子操作
3. 理解量子态和算子的矩阵表示

 **讲师：** "除此之外，还有一些其他内容我们将通过门来实现。但我希望你们真正理解什么是矩阵，我们可以对矩阵应用什么样的算子，以及我们可以通过矩阵表示什么，诸如此类的内容。"

Quantum Computing Lecture - Session Summary (Continuation)

Course Administration & Logistics

HOMEWORK POSTING REQUEST

 **Student:** "Sir, can you please post today's and the other day's homework on our teams so that we can just—"

 **Lecturer:** "Classwork, do you know the homework? Anyone who knows the homework, please post them."

 **Student:** "Thank you, Opa."

 **Lecturer:** "Thank you."

Core Academic Content: Matrix Operations & Quantum Gates

INTRODUCTION TO MATRIX FUNDAMENTALS

The lecturer emphasizes the importance of understanding **matrix operations** before proceeding to quantum gates:

- **Matrix multiplication** and **division** are foundational operations
- Understanding **what a matrix is** and its properties
- Identifying **what operators** can be applied to matrices
- Recognizing **what can be represented** through matrix notation

UPCOMING TOPICS PREVIEW

The next session will cover:

1. **Matrix operations** (multiplication, division)
2. **Gate-based quantum operations**
3. Understanding **matrix representations** of quantum states and operators

 **Lecturer:** "So read something about matrix. So apart from this, there will be some other stuff which we will do through gates. But I want you to actually understand what the matrix is, what sort of an operators we can apply on matrix, what we can represent through matrix, that kind of stuff."

Summary

ADMINISTRATIVE NOTIFICATIONS

- **Homework posting:** Students requested homework to be posted on Microsoft Teams; the lecturer asked students who know the homework to share it publicly
- **Gratitude:** Both lecturer and student exchanged thanks (Opa acknowledged)

CORE ACADEMIC KNOWLEDGE POINTS

1. **Matrix fundamentals** are essential prerequisites for understanding quantum gates
2. **Key matrix operations** to master:
 - Multiplication
 - Division

3. Conceptual understanding required:

- What a matrix represents
- Types of operators applicable to matrices
- Representational capabilities of matrices

LEVEL 9 RECOMMENDED LEARNING RESOURCES

Textbooks:

- *Quantum Computation and Quantum Information* by Nielsen & Chuang (Chapters 2-4 for linear algebra foundations)
- *Mathematics of Quantum Computing* by Wolfgang Scherer (Chapter 1: Linear Algebra Review)

Academic Papers:

- "Quantum Gates and Circuits" – A foundational review paper on gate-based quantum computing
- "Matrix Representations in Quantum Mechanics" – Standard reference for understanding operator formalism

Online Resources:

- MIT OpenCourseWare: *Linear Algebra* (18.06) – Professor Gilbert Strang's lectures on matrix theory
- Qiskit Textbook: *The Atoms of Computation* – Practical introduction to quantum gates and their matrix representations

中文全文翻译

量子计算课程 - 课堂摘要 (续)

课程行政与后勤

作业发布请求

👤 学生: "老师, 您能把今天和之前那天的作业发到我们的Teams上吗? 这样我们就可以——"

👤 讲师: "课堂作业, 你们谁知道作业是什么? 知道作业的同学, 请帮忙发一下。"

👤 学生: "谢谢, Opa。"

👤 讲师: "谢谢。"

核心学术内容: 矩阵运算与量子门

矩阵基础入门

讲师强调了在学习量子门之前理解**矩阵运算**的重要性:

- 矩阵乘法和除法是基础运算
- 理解矩阵是什么及其性质
- 识别可以对矩阵应用哪些算子
- 认识可以通过矩阵表示什么

即将讨论的主题预览

下节课将涵盖:

1. 矩阵运算 (乘法、除法)
2. 基于门的量子操作
3. 理解量子态和算子的矩阵表示

👤 讲师: "所以请阅读一些关于矩阵的内容。除此之外, 还有一些其他内容我们将通过门来实现。但我希望你们真正理解什么是矩阵, 我们可以对矩阵应用什么样的算子, 以及我们可以通过矩阵表示什么, 诸如此类的内容。"

总结

行政通知汇总

- 作业发布：学生请求将作业发布到Microsoft Teams上；讲师请知道作业的同学公开分享
- 致谢：讲师和学生互相表达了感谢（Opa被点名致谢）

核心学术知识点提炼

1. 矩阵基础是理解量子门的必要先决条件

2. 需要掌握的矩阵运算：

- 乘法
- 除法

3. 需要理解的概念：

- 矩阵代表什么
- 可应用于矩阵的算子类型
- 矩阵的表示能力

LEVEL 9 推荐学习资源

教科书：

- 量子计算与量子信息（Nielsen & Chuang著，第2-4章：线性代数基础）
- 量子计算数学（Wolfgang Scherer著，第1章：线性代数复习）

学术论文：

- "量子门与电路"——关于基于门的量子计算的基础综述论文
- "量子力学中的矩阵表示"——理解算子形式体系的标准参考文献

在线资源：

- MIT开放式课程：线性代数（18.06）——Gilbert Strang教授的矩阵理论讲座
- Qiskit教科书：计算的原子——量子门及其矩阵表示的实践入门